Indices

Notes · Questions · Worked Answers

Index notation · Laws of indices · Indices with algebra
Zero & negative indices · Fractional indices



No calculator Calculator allowed

Index

= no calculator = calculator allowed

Sources: **PP** = Past Paper Penguin, *Index Laws* (Cambridge IGCSE 0580 questions), page · question; **RR** = Rocket Revise (0580 questions compiled by Ahmed Gul), page · question; **MEP** = MEP Pupil Text 1 exercises, page · question; **Exam A** = 'Indices with algebra combined' questions; **Exam B** = zero, negative & fractional index questions; **Exam C** = typed exam questions; **Practice** = extra questions written for this booklet. The sources give no exam years. Questions repeated across sources appear only once.

Colour key: 1 Powers, Roots & Index Notation 2 Laws of Indices with Numbers 3 Indices with Algebra 4 Zero & Negative Indices 5 Fractional Indices 6 Negative & Fractional Indices with Algebra 7 Index Equations (Finding the Unknown Power)

1. Powers, Roots & Index Notation

1. PP p.1 · Q1 ..... p. 8
2. RR p.1 · Q2 ..... p. 8
3. MEP · p.4 Q1 ..... p. 8
4. MEP · p.4 Q2 ..... p. 8
5. MEP · p.4 Q3 ..... p. 10
6. MEP · p.5 Q4 ..... p. 10
7. MEP · p.5 Q5(a)–(h) ..... p. 10
8. MEP · p.5 Q5(i)–(p) ..... p. 12
9. MEP · p.5 Q6 ..... p. 12
10. MEP · p.7 Q1 ..... p. 14
11. MEP · p.7 Q2 ..... p. 14

2. Laws of Indices with Numbers

12. PP p.4 · Q3 ..... p. 16
13. Exam A · Q11 ..... p. 16
14. Exam A · Q14 ..... p. 18
15. Exam A · Q16 ..... p. 18
16. MEP · p.7 Q3 ..... p. 20
17. MEP · p.7 Q4 ..... p. 20
18. MEP · p.7 Q5 ..... p. 22
19. MEP · p.8 Q6 ..... p. 22
20. MEP · p.8 Q7 ..... p. 24
21. MEP · p.8 Q8 ..... p. 24
22. MEP · p.8 Q9 ..... p. 24

3. Indices with Algebra

23. PP p.1 · Q5 ..... p. 26
24. PP p.1 · Q6 ..... p. 26
25. PP p.2 · Q4 ..... p. 26
26. PP p.3 · Q10 ..... p. 26
27. PP p.3 · Q11 ..... p. 28
28. PP p.3 · Q9 ..... p. 28
29. PP p.3 · Q13 ..... p. 28
30. PP p.3 · Q7 ..... p. 28
31. PP p.4 · Q10 ..... p. 28
32. PP p.4 · Q7 ..... p. 30
33. PP p.5 · Q9 ..... p. 30
34. RR p.5 · Q6 ..... p. 30
35. RR p.9 · Q13 ..... p. 30
36. RR p.7 · Q1 ..... p. 30
37. RR p.7 · Q5 ..... p. 32
38. RR p.10 · Q16 ..... p. 32
39. Exam A · Q1 ..... p. 32
40. Exam A · Q2 ..... p. 34
41. Exam A · Q3 ..... p. 34
42. Exam A · Q4 ..... p. 34
43. Exam A · Q5 ..... p. 36
44. Exam A · Q6 ..... p. 36
45. Exam A · Q7 ..... p. 38
46. Exam A · Q9 ..... p. 38
47. Exam A · Q10 ..... p. 38
48. Exam A · Q12 ..... p. 40
49. Exam A · Q15 ..... p. 40
50. Exam A · Q17 ..... p. 40
51. Exam A · Q18(i) ..... p. 42

52. Exam C · Q1 ..... p. 42
53. Exam C · Q2 ..... p. 42
54. Exam C · Q3 ..... p. 44
55. Exam C · Q4 ..... p. 44
56. Exam C · Q5 ..... p. 44
57. Exam C · Q7 ..... p. 46
58. Exam C · Q8 ..... p. 46
59. Exam C · Q9 ..... p. 46
60. MEP · p.8 Q10 ..... p. 48

4. Zero & Negative Indices

61. PP p.1 · Q5 ..... p. 50
62. PP p.2 · Q4 ..... p. 50
63. PP p.2 · Q6 ..... p. 50
64. PP p.2 · Q3 ..... p. 50
65. PP p.10 · Q13 ..... p. 52
66. PP p.11 · Q8 ..... p. 52
67. RR p.9 · Q20 ..... p. 52
68. RR p.9 · Q2 ..... p. 52
69. RR p.5 · Q7 ..... p. 54
70. RR p.3 · Q14 ..... p. 54
71. RR p.10 · Q17 ..... p. 54
72. RR p.8 · Q14 ..... p. 54
73. Exam A · Q8 ..... p. 56
74. Exam A · Q13 ..... p. 56
75. Exam B · Q2 ..... p. 58
76. Exam B · Q3 ..... p. 58
77. Exam B · Q4 ..... p. 58
78. Exam B · Q5 ..... p. 60
79. Exam B · Q12 ..... p. 60
80. Exam C · Q6 ..... p. 60

5. Fractional Indices

81. PP p.5 · Q5 ..... p. 62
82. PP p.6 · Q12 ..... p. 62
83. PP p.6 · Q5 ..... p. 62
84. PP p.11 · Q8 ..... p. 62
85. PP p.11 · Q11 ..... p. 62
86. PP p.11 · Q19 ..... p. 64
87. PP p.12 · Q17 ..... p. 64
88. RR p.6 · Q10 ..... p. 64
89. RR p.6 · Q16 ..... p. 66
90. RR p.10 · Q5 ..... p. 66
91. RR p.10 · Q11 ..... p. 66
92. RR p.2 · Q10 ..... p. 66
93. RR p.3 · Q5 ..... p. 68
94. Exam A · Q19 ..... p. 68
95. Exam A · Q20 ..... p. 68
96. Exam B · Q6 ..... p. 70
97. Exam B · Q7 ..... p. 70
98. Exam B · Q9 ..... p. 70
99. Exam B · Q10 ..... p. 72
100. Exam B · Q11(b) ..... p. 72
101. Exam B · Q16 ..... p. 74
102. MEP · p.14 Q1 ..... p. 74
103. MEP · p.14 Q2 ..... p. 76
104. MEP · p.15 Q3 ..... p. 76
105. Practice ..... p. 78

6. Negative & Fractional Indices with Algebra

106. PP p.7 · Q15 ..... p. 80
107. PP p.7 · Q18 ..... p. 80
108. PP p.7 · Q17 ..... p. 80
109. PP p.7 · Q22 ..... p. 80
110. PP p.7 · Q14 ..... p. 80
111. PP p.8 · Q17 ..... p. 82
112. PP p.8 · Q21 ..... p. 82
113. PP p.8 · Q21 ..... p. 82
114. PP p.8 · Q7 ..... p. 82
115. PP p.9 · Q23 ..... p. 84
116. PP p.13 · Q20 ..... p. 84
117. PP p.13 · Q2 ..... p. 84
118. PP p.14 · Q9 ..... p. 86

119. RR p.1 · Q14 ..... p. 86
120. RR p.1 · Q5 ..... p. 86
121. RR p.2 · Q11 ..... p. 88
122. RR p.2 · Q11 ..... p. 88
123. RR p.3 · Q13 ..... p. 88
124. RR p.3 · Q14 ..... p. 88
125. RR p.4 · Q17 ..... p. 90
126. RR p.5 · Q11 ..... p. 90
127. RR p.6 · Q13 ..... p. 90
128. RR p.6 · Q12 ..... p. 92
129. RR p.7 · Q17 ..... p. 92
130. RR p.7 · Q6 ..... p. 92
131. RR p.7 · Q7 ..... p. 92
132. RR p.7 · Q10 ..... p. 92
133. RR p.8 · Q25 ..... p. 94
134. RR p.8 · Q8 ..... p. 94
135. RR p.9 · Q27 ..... p. 94
136. RR p.11 · Q27 ..... p. 96
137. Exam B · Q13 ..... p. 96
138. Exam C · Q10 ..... p. 96
139. Exam C · Q11 ..... p. 98
140. MEP · p.15 Q4 ..... p. 98
141. Practice ..... p. 100

7. Index Equations (Finding the Unknown Power)

142. PP p.1 · Q11 ..... p. 102
143. PP p.2 · Q10 ..... p. 102
144. PP p.10 · Q15 ..... p. 102
145. PP p.9 · Q26 ..... p. 102
146. PP p.9 · Q21 ..... p. 104
147. PP p.15 · Q13 ..... p. 104
148. PP p.15 · Q6 ..... p. 104
149. PP p.16 · Q11 ..... p. 106
150. PP p.16 · Q17 ..... p. 106
151. PP p.17 · Q16 ..... p. 106
152. PP p.17 · Q13 ..... p. 106
153. PP p.17 · Q18 ..... p. 108
154. PP p.18 · Q25 ..... p. 108
155. PP p.18 · Q(d) ..... p. 108
156. PP p.19 · Q22 ..... p. 110
157. PP p.20 · Q21 ..... p. 110
158. PP p.21 · Q(e) ..... p. 110
159. RR p.1 · Q8 ..... p. 112
160. RR p.2 · Q14 ..... p. 112
161. RR p.5 · Q3 ..... p. 112
162. RR p.8 · Q6 ..... p. 112
163. RR p.10 · Q9 ..... p. 114
164. RR p.4 · Q16 ..... p. 114
165. Practice ..... p. 114

Key Notes

1. Index notation, squares, cubes and roots

a^n means a multiplied by itself n times:

$5^3 = 5 \times 5 \times 5 = 125, \quad 2^7 = 128.$

Square root: $\sqrt{49} = 7$ because $7^2 = 49$.

Cube root: $\sqrt[3]{64} = 4$ because $4^3 = 64$.

Work out powers **before** adding: $6^2 + 8^2 = 36 + 64 = 100$.

n	n^2	n^3
2	4	8
3	9	27
4	16	64
5	25	125
6	36	216
10	100	1000
11	121	1331
12	144	1728
13	169	2197
15	225	3375

2. The laws of indices (same base only!)

Multiply → add	$a^m \times a^n = a^{m+n}$	$3^4 \times 3^5 = 3^9$
Divide → subtract	$a^m \div a^n = a^{m-n}$	$7^8 \div 7^3 = 7^5$
Power of a power → multiply	$(a^m)^n = a^{mn}$	$(2^3)^4 = 2^{12}$
Zero index	$a^0 = 1$	$5^0 = 1, \ x^0 = 1$
Index 1	$a^1 = a$	$2 \times 2^7 = 2^1 \times 2^7 = 2^8$

Careful: $2^3 \times 3^2$ cannot be combined (different bases). $3 \times 2^3 = 3 \times 8 = 24$, **not** 6^3.

To use the laws, write numbers as powers of the same base: $4 \times 2^2 = 2^2 \times 2^2 = 2^4$, $\dfrac{2^6}{8} = \dfrac{2^6}{2^3} = 2^3$.

3. Indices with algebra

Numbers (coefficients) are multiplied or divided normally; indices of each letter are added or subtracted.

$5x^4y^3 \times x^2y = 5x^6y^4$ $\qquad (y = y^1)$

$\dfrac{45e^6f^8}{5ef^2} = 9e^5f^6$ $\qquad (45 \div 5 = 9;\ e^{6-1};\ f^{8-2})$

$(2n^3)^4 = 2^4n^{12} = 16n^{12}$ $\qquad$ **everything** in the bracket is raised to the power

$(16y^6)^{\frac{1}{2}} = 4y^3$ $\qquad 16^{\frac{1}{2}} = 4$ and $6 \times \frac{1}{2} = 3$

Common errors: $w^4 \times w^4 = w^8$ (not w^{16}); $(2m^4)^3 = 8m^{12}$ (not $6m^{12}$ or $2m^{12}$).

4. Negative indices $\rightarrow$ reciprocal

$$a^{-n} = \frac{1}{a^n} \qquad 4^{-2} = \frac{1}{4^2} = \frac{1}{16}, \qquad 2^{-1} = \frac{1}{2}, \qquad \frac{1}{p^2} = p^{-2}$$

- A negative index does **not** make the answer negative: $3^{-2} = \frac{1}{9}$, not -9.
- Fraction with a negative index: **flip it** and make the index positive: $\left(\frac{1}{4}\right)^{-1} = 4$, $\left(\frac{2}{5}\right)^{-2} = \left(\frac{5}{2}\right)^2 = \frac{25}{4}$.
- "No negative indices" means move the term to the other side of the fraction line: $4y^{-2} = \dfrac{4}{y^2}$, $(2t^{-3})^{-2} = \dfrac{t^6}{4}$.
- Subtracting a negative: $a^4 \div a^{-2} = a^{4-(-2)} = a^6$.

5. Fractional indices $\rightarrow$ roots

$$a^{\frac{1}{n}} = \sqrt[n]{a} \qquad a^{\frac{m}{n}} = \left(\sqrt[n]{a}\right)^m$$

Denominator = root, numerator = power. Do the **root first** (smaller numbers):

$$9^{\frac{1}{2}} = 3, \quad 27^{\frac{1}{3}} = 3, \quad 16^{\frac{3}{4}} = (\sqrt[4]{16})^3 = 2^3 = 8, \quad 25^{\frac{3}{2}} = (\sqrt{25})^3 = 125$$

Negative and fractional together: reciprocal, then root, then power:

$$64^{-\frac{2}{3}} = \frac{1}{(\sqrt[3]{64})^2} = \frac{1}{4^2} = \frac{1}{16} \qquad \left(\frac{16}{81}\right)^{-\frac{3}{4}} = \left(\frac{81}{16}\right)^{\frac{3}{4}} = \left(\frac{3}{2}\right)^3 = \frac{27}{8}$$

Roots as indices: $\sqrt{m} = m^{\frac{1}{2}}$, $\sqrt[3]{q^2} = q^{\frac{2}{3}}$, $8\sqrt{8} = 8^1 \times 8^{\frac{1}{2}} = 8^{\frac{3}{2}}$.

6. Solving index equations

Write both sides as powers of the **same base**, then equate the indices.

$3^n = \frac{1}{9} = 3^{-2} \Rightarrow n = -2$; $\quad 9^x = 27 \Rightarrow 3^{2x} = 3^3 \Rightarrow x = \frac{3}{2}$; $\quad 7^5 \times 7^6 = 7^3 \times 7^k \Rightarrow 11 = 3 + k \Rightarrow k = 8$.

7. TI-Nspire CX: powers and roots

- Power: [^], square: [x^2]. Negative sign: the [(-)] key, e.g. `8^(-1)`.
- Fractional index: **always use brackets**: `81^(3/2)` $= 729$ (`81^3/2` gives 265 720.5 — wrong!).
- [ctrl] [x^2] $= \sqrt{\ }$; [ctrl] [^] $= \sqrt[n]{\ }$ template. [ctrl] [enter] gives a decimal answer.

1

No calculator

PP p.1 · Q1

[Maximum mark: 1]

Work out. 2^4

2

No calculator

RR p.1 · Q2

[Maximum mark: 2]

Show that $\left(\frac{1}{10}\right)^2 + \left(\frac{2}{5}\right)^2 = 0.17$.

Write down all the steps in your working.

3

No calculator

MEP Pupil Text 1 · p.4 Q1

[Maximum mark: 8]

Find

(a) 5^2 **(b)** 6^2 **(c)** 1^2 **(d)** 7^2

Use your answers to find

(e) $\sqrt{36}$ **(f)** $\sqrt{1}$ **(g)** $\sqrt{49}$ **(h)** $\sqrt{25}$

4

No calculator

MEP Pupil Text 1 · p.4 Q2

[Maximum mark: 8]

Find

(a) 3^3 **(b)** 4^3 **(c)** 6^3 **(d)** 10^3

Use your answers to find

(e) $\sqrt[3]{27}$ **(f)** $\sqrt[3]{1000}$ **(g)** $\sqrt[3]{216}$ **(h)** $\sqrt[3]{64}$

1 *PP p.1 · Q1*

$2 \times 2 \times 2 \times 2 = \boxed{16}$ (not 8) **B1**

2 *RR p.1 · Q2*

$\frac{1}{100} + \frac{4}{25}$ **M1**

$= \frac{1}{100} + \frac{16}{100} = \frac{17}{100} = 0.17$ **A1**

3 *MEP Pupil Text 1 · p.4 Q1*

(a) $5^2 = 25$ **(b)** $6^2 = 36$ **(c)** $1^2 = 1$ **(d)** $7^2 = 49$

(e) $\sqrt{36} = 6$ **(f)** $\sqrt{1} = 1$ **(g)** $\sqrt{49} = 7$ **(h)** $\sqrt{25} = 5$

4 *MEP Pupil Text 1 · p.4 Q2*

(a) $3^3 = 27$ **(b)** $4^3 = 64$ **(c)** $6^3 = 216$ **(d)** $10^3 = 1000$

(e) $\sqrt[3]{27} = 3$ **(f)** $\sqrt[3]{1000} = 10$ **(g)** $\sqrt[3]{216} = 6$ **(h)** $\sqrt[3]{64} = 4$

5

No calculator

MEP Pupil Text 1 · p.4 Q3

[Maximum mark: 12]

Find

(a) 10^2 **(b)** 2^2 **(c)** 4^2 **(d)** 7^2

(e) 8^2 **(f)** 9^2 **(g)** 1^3 **(h)** 7^3

(i) 8^3 **(j)** 0^2 **(k)** 0^3 **(l)** 2^3

6

No calculator

MEP Pupil Text 1 · p.5 Q4

[Maximum mark: 6]

Find

(a) $\sqrt{100}$ **(b)** $\sqrt{4}$ **(c)** $\sqrt{81}$ **(d)** $\sqrt{64}$

(e) $\sqrt{16}$ **(f)** $\sqrt{9}$

7

Calculator

MEP Pupil Text 1 · p.5 Q5(a)–(h)

[Maximum mark: 8]

Use a calculator to find

(a) 12^2 **(b)** 11^2 **(c)** 15^3 **(d)** 13^3

(e) 13^2 **(f)** 15^2 **(g)** 20^2 **(h)** 11^3

5 *MEP Pupil Text 1 · p.4 Q3*

(a) $10^2 = 100$ **(b)** $2^2 = 4$ **(c)** $4^2 = 16$ **(d)** $7^2 = 49$

(e) $8^2 = 64$ **(f)** $9^2 = 81$ **(g)** $1^3 = 1$ **(h)** $7^3 = 343$

(i) $8^3 = 512$ **(j)** $0^2 = 0$ **(k)** $0^3 = 0$ **(l)** $2^3 = 8$

Remember: $a^3 = a \times a \times a$, e.g. $7^3 = 7 \times 7 \times 7 = 343$; 0 to any positive power is 0.

6 *MEP Pupil Text 1 · p.5 Q4*

(a) $\sqrt{100} = 10$ **(b)** $\sqrt{4} = 2$ **(c)** $\sqrt{81} = 9$

(d) $\sqrt{64} = 8$ **(e)** $\sqrt{16} = 4$ **(f)** $\sqrt{9} = 3$

7 *MEP Pupil Text 1 · p.5 Q5(a)–(h)*

(a) $12^2 = 144$ **(b)** $11^2 = 121$ **(c)** $15^3 = 3375$ **(d)** $13^3 = 2197$

(e) $13^2 = 169$ **(f)** $15^2 = 225$ **(g)** $20^2 = 400$ **(h)** $11^3 = 1331$

Learn these — they are the answers you need for the roots in Q5(i)–(p).

TI-Nspire CX: Question 7

Type 12 [x^2] [enter] → 144. For cubes type 15 [^] 3 [enter] → 3375.

8

No calculator

MEP Pupil Text 1 · p.5 Q5(i)–(p)

[Maximum mark: 8]

Without a calculator, find

(i) $\sqrt{121}$ **(j)** $\sqrt{400}$ **(k)** $\sqrt{169}$ **(l)** $\sqrt{225}$

(m) $\sqrt[3]{3375}$ **(n)** $\sqrt[3]{2197}$ **(o)** $\sqrt{144}$ **(p)** $\sqrt[3]{1331}$

9

No calculator

MEP Pupil Text 1 · p.5 Q6

[Maximum mark: 8]

Find

(a) 6^2+4^2 **(b)** 3^2-2^2 **(c)** 10^2+4^2 **(d)** 3^2+4^2

(e) 5^2-3^2 **(f)** 4^3+2^3 **(g)** 1^3+10^3 **(h)** 6^2+8^2

8 *MEP Pupil Text 1 · p.5 Q5(i)–(p)*

(i) $\sqrt{121} = 11$ **(j)** $\sqrt{400} = 20$ **(k)** $\sqrt{169} = 13$ **(l)** $\sqrt{225} = 15$

(m) $\sqrt[3]{3375} =$ **(n)** $\sqrt[3]{2197} = 13$ **(o)** $\sqrt{144} = 12$ **(p)** $\sqrt[3]{1331} = 11$

15

Use Q5(a)–(h): e.g. $15^3 = 3375$ so $\sqrt[3]{3375} = 15$.

9 *MEP Pupil Text 1 · p.5 Q6*

(a) $6^2 + 4^2 = 36 + 16 = 52$

(b) $3^2 - 2^2 = 9 - 4 = 5$

(c) $10^2 + 4^2 = 100 + 16 = 116$

(d) $3^2 + 4^2 = 9 + 16 = 25$

(e) $5^2 - 3^2 = 25 - 9 = 16$

(f) $4^3 + 2^3 = 64 + 8 = 72$

(g) $1^3 + 10^3 = 1 + 1000 = 1001$

(h) $6^2 + 8^2 = 36 + 64 = 100$

Work out each power first, then add or subtract. $6^2 + 4^2 \neq 10^2$!

10

No calculator

MEP Pupil Text 1 · p.7 Q1

[Maximum mark: 10]

Write each of the following using index notation.

(a) $4 \times 4 \times 4 \times 4 \times 4$

(b) $3 \times 3 \times 3$

(c) $6 \times 6 \times 6 \times 6 \times 6 \times 6 \times 6$

(d) $7 \times 7 \times 7 \times 7$

(e) $18 \times 18 \times 18$

(f) 19×19

(g) $4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4$

(h) $7 \times 7 \times 7 \times 7 \times 7$

(i) $10 \times 10 \times 10 \times 10 \times 10 \times 10$

(j) $100 \times 100 \times 100 \times 100 \times 100$

11

No calculator

MEP Pupil Text 1 · p.7 Q2

[Maximum mark: 12]

Find the value of each of the following.

(a) 3^4

(b) 5^4

(c) 7^4

(d) 10^4

(e) 5^0

(f) 3^6

(g) 2^7

(h) 2^1

(i) 8^4

(j) 4^1

(k) 3^0

(l) 5^2

10 *MEP Pupil Text 1 · p.7 Q1*

(a) $4 \times 4 \times 4 \times 4 \times 4 = 4^5$

(b) $3 \times 3 \times 3 = 3^3$

(c) $6 \times \cdots \times 6$ (7 sixes) $= 6^7$

(d) $7 \times 7 \times 7 \times 7 = 7^4$

(e) $18 \times 18 \times 18 = 18^3$

(f) $19 \times 19 = 19^2$

(g) $4 \times \cdots \times 4$ (7 fours) $= 4^7$

(h) $7 \times \cdots \times 7$ (5 sevens) $= 7^5$

(i) $10 \times \cdots \times 10$ (6 tens) $= 10^6$

(j) $100 \times \cdots \times 100$ (5) $= 100^5$

The index (power) counts how many times the base is multiplied.

11 *MEP Pupil Text 1 · p.7 Q2*

(a) $3^4 = 81$

(b) $5^4 = 625$

(c) $7^4 = 2401$

(d) $10^4 = 10000$

(e) $5^0 = \mathbf{1}$

(f) $3^6 = 729$

(g) $2^7 = 128$

(h) $2^1 = 2$

(i) $8^4 = 4096$

(j) $4^1 = 4$

(k) $3^0 = \mathbf{1}$

(l) $5^2 = 25$

Any number to the power 0 is 1 ($a^0 = 1$); any number to the power 1 is itself.

12

No calculator

PP p.4 · Q3

[Maximum mark: 6]

(a) Simplify, giving your answer as a single power of 7.

(i) $7^5 \times 7^6$ [1]

(ii) $7^{15} \div 7^5$ [1]

(iii) $42 + 7$ [1]

(b) Simplify. $(5x^2 \times 2xy^4)^3$ [3]

13

No calculator

Exam A · Q11

[Maximum mark: 2]

$7^5 \times 7^6 = 7^3 \times 7^k$

Find the value of k.

12 *PP p.4 · Q3*

(a)(i) $\boxed{7^{11}}$ **B1** (ii) $\boxed{7^{10}}$ **B1** (iii) $49 = \boxed{7^2}$ **B1**

(b) $(10x^3y^4)^3$ **M1** $= 10^3x^9y^{12} = \boxed{1000x^9y^{12}}$ **A2**

13 *Exam A · Q11*

$7^{11} = 7^{3+k}$ **M1**

$3 + k = 11 \Rightarrow \boxed{k = 8}$ **A1**

14

No calculator

Exam A · Q14

[Maximum mark: 4]

(a) Write as a power of 7

(i) $7^8 \div 7^3$ (ii) $\dfrac{7^2 \times 7^3}{7}$ [3]

(b) Write down the reciprocal of 2. [1]

15

No calculator

Exam A · Q16

[Maximum mark: 3]

Work out the value of

(i) 4^2 (ii) $\sqrt{64}$ (iii) 3×2^3

14 *Exam A · Q14*

(a)(i) $7^{8-3} = \boxed{7^5}$ **B1**

(ii) $\dfrac{7^5}{7^1} = \boxed{7^4}$ **M1 A1**

(b) $\boxed{\dfrac{1}{2}}$ (or 0.5, 2^{-1}) **B1**

15 *Exam A · Q16*

(i) $\boxed{16}$ **B1**

(ii) $\boxed{8}$ **B1**

(iii) $3 \times 8 = \boxed{24}$ (not 6^3!) **B1**

16

No calculator

MEP Pupil Text 1 · p.7 Q3

[Maximum mark: 21]

Fill in the missing numbers.

(a) $2^7 \times 2^4 = 2^?$ **(b)** $3^4 \times 3^5 = 3^?$ **(c)** $3^6 \times 3^7 = 3^?$

(d) $4^? \times 4^2 = 4^7$ **(e)** $5^? \times 5^2 = 5^6$ **(f)** $5^4 \times 5^? = 5^9$

(g) $?^2 \times 4^4 = 4^6$ **(h)** $5^7 \div 5^4 = 5^?$ **(i)** $3^4 \div 3^2 = 3^?$

(j) $7^{14} \div 7^{10} = 7^?$ **(k)** $17^5 \div 17^? = 17^3$ **(l)** $9^7 \div 9^? = 9^3$

(m) $4^6 \times 4^? = 4^{11}$ **(n)** $4^? \div 4^6 = 4^{10}$ **(o)** $3^? \times 3^2 = 3^8$

(p) $3^6 \div 3^6 = ?$ **(q)** $3^7 \div 3^6 = ?$ **(r)** $3^0 \times 3^? = 3^5$

(s) $3^0 \times 3^7 = 3^?$ **(t)** $4^1 \times 4^? = 4^8$ **(u)** $5^2 \times 5^? = 5^2$

17

No calculator

MEP Pupil Text 1 · p.7 Q4

[Maximum mark: 9]

Fill in the missing numbers.

(a) $4 = 2^?$ **(b)** $8 = 2^?$ **(c)** $16 = 2^?$

(d) $64 = 2^?$ **(e)** $27 = 3^?$ **(f)** $25 = 5^?$

(g) $64 = 4^?$ **(h)** $81 = 3^?$ **(i)** $125 = ?^3$

16 *MEP Pupil Text 1 · p.7 Q3*

Multiply $\Rightarrow$ **add** the indices; **divide** $\Rightarrow$ **subtract** the indices.

(a) 11 **(b)** 9 **(c)** 13 **(d)** 5

(e) 4 **(f)** 5 **(g)** 4 **(h)** 3

(i) 2 **(j)** 4 **(k)** 2 **(l)** 4

(m) 5 **(n)** 16 **(o)** 6 **(p)** $3^0 = 1$

(q) $3^1 = 3$ **(r)** 5 **(s)** 7 **(t)** 7

(u) 0

e.g. (n) $4^? \div 4^6 = 4^{10} \Rightarrow ? - 6 = 10 \Rightarrow ? = 16$; (u) $2 + ? = 2 \Rightarrow ? = 0$.

17 *MEP Pupil Text 1 · p.7 Q4*

(a) $4 = 2^{\mathbf{2}}$ **(b)** $8 = 2^{\mathbf{3}}$ **(c)** $16 = 2^{\mathbf{4}}$

(d) $64 = 2^{\mathbf{6}}$ **(e)** $27 = 3^{\mathbf{3}}$ **(f)** $25 = 5^{\mathbf{2}}$

(g) $64 = 4^{\mathbf{3}}$ **(h)** $81 = 3^{\mathbf{4}}$ **(i)** $125 = \mathbf{5}^3$

18

No calculator

MEP Pupil Text 1 · p.7 Q5

[Maximum mark: 15]

Simplify the following expressions, giving your answer in index notation.

(a) $3^7 \times 3^6$ **(b)** 2×2^7 **(c)** $4^5 \times 4^6$

(d) $3^6 \times 3^4$ **(e)** $2^4 \times 2^5$ **(f)** $2^6 \times 2^4$

(g) $3^7 \div 3^2$ **(h)** 3×3^6 **(i)** $3^6 \div 3$

(j) $\dfrac{8^{12}}{8^2}$ **(k)** $\dfrac{7^6}{7^3}$ **(l)** $\dfrac{9^2}{9^0}$

(m) 4×2^2 **(n)** $\dfrac{2^5}{4}$ **(o)** $\dfrac{2^6}{8}$

19

No calculator

MEP Pupil Text 1 · p.8 Q6

[Maximum mark: 12]

Fill in the missing powers.

(a) $8 = 2^{?}$ **(b)** $1000 = 10^{?}$ **(c)** $16 = 2^{?}$

(d) $27 = 3^{?}$ **(e)** $81 = 3^{?}$ **(f)** $10\,000 = 10^{?}$

(g) $625 = 5^{?}$ **(h)** $64 = 4^{?}$ **(i)** $1296 = 6^{?}$

(j) $1 = 2^{?}$ **(k)** $36 = 6^{?}$ **(l)** $1 = 5^{?}$

18 *MEP Pupil Text 1 · p.7 Q5*

(a) 3^{13} **(b)** 2^{8} **(c)** 4^{11} **(d)** 3^{10}

(e) 2^{9} **(f)** 2^{10} **(g)** 3^{5} **(h)** 3^{7}

(i) 3^{5} **(j)** 8^{10} **(k)** 7^{3} **(l)** 9^{2}

(m) $2^2 \times 2^2 = 2^4$ **(n)** $2^5 \div 2^2 = 2^3$ **(o)** $2^6 \div 2^3 = 2^3$

Note: $2 = 2^1$ and $3 = 3^1$, so $2 \times 2^7 = 2^{1+7}$. For (m)–(o) first write $4 = 2^2$, $8 = 2^3$.

19 *MEP Pupil Text 1 · p.8 Q6*

(a) 3 **(b)** 3 **(c)** 4 **(d)** 3

(e) 4 **(f)** 4 **(g)** 4 **(h)** 3

(i) 4 **(j)** 0 **(k)** 2 **(l)** 0

(j), (l): $1 = a^0$ for any base a.

20

No calculator

MEP Pupil Text 1 · p.8 Q7

[Maximum mark: 9]

Simplify the following, giving your answers in index form.

(a) $(2^3)^2$ **(b)** $(3^2)^2$ **(c)** $(6^2)^3$

(d) $(5^3)^2$ **(e)** $(2^2)^4$ **(f)** $(4^2)^3$

(g) $(3^2)^4$ **(h)** $(5^2)^4$ **(i)** $(3^3)^2$

21

No calculator

MEP Pupil Text 1 · p.8 Q8

[Maximum mark: 6]

Fill in the missing numbers.

(a) $(2^2)^4 = 2^?$ **(b)** $(2^?)^3 = 2^{12}$ **(c)** $(3^2)^5 = ?^{10}$

(d) $(5^?)^4 = 5^{12}$ **(e)** $(10^5)^? = 10^{15}$ **(f)** $(7^5)^? = 7^{20}$

22

No calculator

MEP Pupil Text 1 · p.8 Q9

[Maximum mark: 9]

Simplify each of the following, giving your answer in index notation.

(a) $3^2 \times 3^0 \times 3^4$ **(b)** $2^6 \times 2^7 \times 2$ **(c)** $5^2 \times 5^7 \times 5^3$

(d) $\dfrac{7^2 \times 7^4}{7^3}$ **(e)** $\dfrac{7^4 \times 7^5}{7^2 \times 7^3}$ **(f)** $\dfrac{2^3 \times 2^8}{2^3 \times 2}$

(g) $\dfrac{3^2 \times 3^3}{3^5}$ **(h)** $\dfrac{4^7 \times 4^8}{4^5 \times 4^9}$ **(i)** $\dfrac{2^3 \times 2^0}{2^2}$

20 *MEP Pupil Text 1 · p.8 Q7*

Power of a power ⇒ **multiply** the indices: $(a^m)^n = a^{mn}$.

(a) $(2^3)^2 = 2^6$ **(b)** $(3^2)^2 = 3^4$ **(c)** $(6^2)^3 = 6^6$

(d) $(5^3)^2 = 5^6$ **(e)** $(2^2)^4 = 2^8$ **(f)** $(4^2)^3 = 4^6$

(g) $(3^2)^4 = 3^8$ **(h)** $(5^2)^4 = 5^8$ **(i)** $(3^3)^2 = 3^6$

21 *MEP Pupil Text 1 · p.8 Q8*

(a) 8 **(b)** 4 **(c)** 3

(d) 3 **(e)** 3 **(f)** 4

e.g. (b) $? \times 3 = 12 \Rightarrow ? = 4$; (c) $(3^2)^5 = 3^{10}$ so the base is 3.

22 *MEP Pupil Text 1 · p.8 Q9*

(a) $3^{2+0+4} = 3^6$ **(b)** $2^{6+7+1} = 2^{14}$ **(c)** $5^{2+7+3} = 5^{12}$

(d) $7^{6-3} = 7^3$ **(e)** $7^{9-5} = 7^4$ **(f)** $2^{11-4} = 2^7$

(g) $3^{5-5} = 3^0 = 1$ **(h)** $4^{15-14} = 4^1 = 4$ **(i)** $2^{3-2} = 2^1 = 2$

23

No calculator PP p.1 · Q5

[Maximum mark: 1]

Simplify $d^8 \div d^2$.

24

No calculator PP p.1 · Q6

[Maximum mark: 3]

Simplify.

(a) $p^2 \times p^4$ [1]

(b) $m^{15} \div m^5$ [1]

(c) $(k^3)^5$ [1]

25

No calculator PP p.2 · Q4

[Maximum mark: 1]

Simplify $\dfrac{a^3 \times a^7}{a^4}$.

26

No calculator PP p.3 · Q10

[Maximum mark: 3]

Simplify.

(a) $n^5 \times n$ [1]

(b) $8x^6 \div 2x^2$ [2]

23 *PP p.1 · Q5*

$\boxed{d^6}$ (subtract: $8-2$; not d^4) **B1**

24 *PP p.1 · Q6*

(a) $\boxed{p^6}$ **B1** (b) $\boxed{m^{10}}$ **B1** (c) $\boxed{k^{15}}$ **B1**

25 *PP p.2 · Q4*

$\dfrac{a^{10}}{a^4} = \boxed{a^6}$ **B1**

26 *PP p.3 · Q10*

(a) $n^5 \times n^1 = \boxed{n^6}$ **B1**

(b) $\boxed{4x^4}$ **B2** (B1 for $4x^k$ or kx^4)

27

No calculator PP p.3 · Q11

[Maximum mark: 2]

Simplify $3x^3 \times 4x^4$.

28

No calculator PP p.3 · Q9

[Maximum mark: 2]

Simplify $4x^4 \times 5x^5$.

29

No calculator PP p.3 · Q13

[Maximum mark: 2]

Simplify $8t^8 \div 4t^4$.

30

No calculator PP p.3 · Q7

[Maximum mark: 2]

Simplify. $30t^{30} \div 5t^5$

31

No calculator PP p.4 · Q10

[Maximum mark: 2]

Simplify $18x^{18} \div 9x^9$.

27 *PP p.3 · Q11*

$\boxed{12x^7}$ **B2**

28 *PP p.3 · Q9*

$\boxed{20x^9}$ **B2**

29 *PP p.3 · Q13*

$\boxed{2t^4}$ (not $2t^2$) **B2**

30 *PP p.3 · Q7*

$\boxed{6t^{25}}$ **B2**

31 *PP p.4 · Q10*

$\boxed{2x^9}$ **B2**

32

No calculator PP p.4 · Q7

[Maximum mark: 2]

Simplify. $32g^{32} \div 4g^4$

33

No calculator PP p.5 · Q9

[Maximum mark: 2]

Simplify $(3w^3)^3$.

34

No calculator RR p.5 · Q6

[Maximum mark: 2]

Simplify. $3x^2y^3 \times x^4y$

35

No calculator RR p.9 · Q13

[Maximum mark: 3]

Simplify.

(a) $(m^5)^2$ [1]

(b) $4x^3y \times 5x^2y$ [2]

36

No calculator RR p.7 · Q1

[Maximum mark: 1]

Simplify. $(x^2)^5$

32 *PP p.4 · Q7*

$\boxed{8g^{28}}$ **B2**

33 *PP p.5 · Q9*

$3^3 \times w^9 = \boxed{27w^9}$ **B2** (not $9w^9$ or $27w^6$)

34 *RR p.5 · Q6*

$\boxed{3x^6y^4}$ **B2**

35 *RR p.9 · Q13*

(a) $\boxed{m^{10}}$ **B1**

(b) $\boxed{20x^5y^2}$ **B2**

36 *RR p.7 · Q1*

$\boxed{x^{10}}$ **B1**

37

No calculator

RR p.7 · Q5

[Maximum mark: 2]

Simplify. $36y^5 \div 4y^2$

38

No calculator

RR p.10 · Q16

[Maximum mark: 1]

(a) Simplify $\frac{w^7}{w^3}$.

(part (b) is the same as PP p.5 Q9 and is not repeated)

39

No calculator

Exam A · Q1

[Maximum mark: 4]

(a) Simplify $m^3 \times m^6$ [1]

(b) Simplify $\frac{p^8}{p^2}$ [1]

(c) Simplify $(2n^3)^4$ [2]

37 *RR p.7 · Q5*

$\boxed{9y^3}$ **B2**

38 *RR p.10 · Q16*

$\boxed{w^4}$ **B1**

39 *Exam A · Q1*

(a) $\boxed{m^9}$ **B1**

(b) $\boxed{p^6}$ **B1**

(c) $2^4 \times n^{12} = \boxed{16n^{12}}$ **B2**

(B1 for $16n^k$ or an^{12})

40

No calculator **Exam A · Q2**

[Maximum mark: 4]

(a) Simplify $m^6 \times m^7$ [1]

(b) Simplify x^0 [1]

(c) Simplify $(16y^6)^{\frac{1}{2}}$ [2]

41

No calculator **Exam A · Q3**

[Maximum mark: 3]

(a) Simplify $m^5 \div m^3$ [1]

(b) Simplify $5x^4y^3 \times x^2y$ [2]

42

No calculator **Exam A · Q4**

[Maximum mark: 4]

(a) Simplify $a^4 \times a^5$ [1]

(b) Simplify $\dfrac{45e^6f^8}{5ef^2}$ [2]

(c) Write down the value of $9^{\frac{1}{2}}$ [1]

40 *Exam A · Q2*

(a) $\boxed{m^{13}}$ **B1**

(b) $\boxed{1}$ **B1**

(c) $16^{\frac{1}{2}} \times y^{6\times\frac{1}{2}} = \boxed{4y^3}$ **B2**

(B1 for $4y^k$ or ay^3)

41 *Exam A · Q3*

(a) $\boxed{m^2}$ **B1**

(b) $5 \times x^{4+2} \times y^{3+1} = \boxed{5x^6y^4}$ **B2**

(B1 for two of 5, x^6, y^4 correct)

42 *Exam A · Q4*

(a) $\boxed{a^9}$ **B1**

(b) $\frac{45}{5}e^{6-1}f^{8-2} = \boxed{9e^5f^6}$ **B2**

(c) $\sqrt{9} = \boxed{3}$ (accept ± 3) **B1**

43

No calculator **Exam A · Q5**

[Maximum mark: 4]

(a) Simplify $m^2 \times m^4$ [1]

(b) Simplify $y^7 \div y^5$ [1]

(c) Simplify $(m^3)^5$ [2]

44

No calculator **Exam A · Q6**

[Maximum mark: 5]

Simplify fully

(a) $p^2 \times p^7$ [1]

(b) $\dfrac{3q^4 \times 2q^5}{q^3}$ [2]

(c) $(2xy^3)^5$ [2]

43 *Exam A · Q5*

(a) $\boxed{m^6}$ **B1**

(b) $\boxed{y^2}$ **B1**

(c) $\boxed{m^{15}}$ **B2**

44 *Exam A · Q6*

(a) $\boxed{p^9}$ **B1**

(b) $\dfrac{6q^9}{q^3} = \boxed{6q^6}$ **M1 A1**

(c) $2^5x^5y^{15} = \boxed{32x^5y^{15}}$ **B2**

45

No calculator **Exam A · Q7**

[Maximum mark: 4]

(a) Simplify $15y^6 \div 3y^2$ [2]

(b) Simplify $7wx^2 \times 3w^3x$ [2]

46

No calculator **Exam A · Q9**

[Maximum mark: 3]

(a) Write down the value of $49^{\frac{1}{2}}$ [1]

(b) Simplify (i) $\dfrac{x^6}{x^2}$ (ii) $(y^4)^3$ [2]

47

No calculator **Exam A · Q10**

[Maximum mark: 2]

Simplify $\dfrac{15a^3b^7}{3a^2b^3}$

45 *Exam A · Q7*

(a) $\boxed{5y^4}$ **B2**

(b) $21\,w^{1+3}x^{2+1} = \boxed{21w^4x^3}$ **B2**

46 *Exam A · Q9*

(a) $\sqrt{49} = \boxed{7}$ **B1**

(b)(i) $\boxed{x^4}$ **B1** (ii) $\boxed{y^{12}}$ **B1**

47 *Exam A · Q10*

$\dfrac{15}{3}a^{3-2}b^{7-3} = \boxed{5ab^4}$ **B2**

(B1 for two of 5, a, b^4)

48

No calculator **Exam A · Q12**

[Maximum mark: 3]

(a) Simplify $p^3 \times p^2$ [1]

(b) Simplify $\dfrac{q^3 \times q^4 \times q}{q^2}$ [2]

49

No calculator **Exam A · Q15**

[Maximum mark: 4]

(a) Simplify $12y^3 \div 3y^5$ [2]

(b) Simplify $2w^3x^2 \times 3w^4x$ [2]

50

No calculator **Exam A · Q17**

[Maximum mark: 6]

Simplify

(i) $x^4 \times x^5$ (ii) $\dfrac{p^8}{p^3}$

(iii) $3s^2t^3 \times 4s^4t^2$ (iv) $(q^3)^4$

48 *Exam A · Q12*

(a) $\boxed{p^5}$ **B1**

(b) $\dfrac{q^8}{q^2} = \boxed{q^6}$ **M1 A1**

49 *Exam A · Q15*

(a) $4y^{3-5} = 4y^{-2} = \boxed{\dfrac{4}{y^2}}$ **B2**

(b) $\boxed{6w^7x^3}$ **B2**

50 *Exam A · Q17*

(i) $\boxed{x^9}$ **B1**

(ii) $\boxed{p^5}$ **B1**

(iii) $\boxed{12s^6t^5}$ **B2**

(iv) $\boxed{q^{12}}$ **B2**

51

No calculator **Exam A · Q18(i)**

[Maximum mark: 1]

Simplify fully $(p^3)^3$

(part (ii) is the same as Q6(b) and is not repeated)

52

No calculator **Exam C · Q1**

[Maximum mark: 3]

Simplify

(a) $m^5 \times m^3$ [1]

(b) $m^8 \div m^2$ [1]

(c) $(m^3)^2$ [1]

53

No calculator **Exam C · Q2**

[Maximum mark: 2]

Simplify $\dfrac{m^9 \times m}{m^5}$

51 *Exam A · Q18(i)*

$p^{3\times 3} = \boxed{p^9}$ **B1**

52 *Exam C · Q1*

(a) $\boxed{m^8}$ **B1** (b) $\boxed{m^6}$ **B1** (c) $\boxed{m^6}$ **B1**

53 *Exam C · Q2*

$\dfrac{m^{10}}{m^5} = \boxed{m^5}$ **M1 A1**

54

No calculator Exam C · Q3

[Maximum mark: 2]

Simplify

(a) $w^4 \times w^4$ [1]

(b) $a^6 \div a^3$ [1]

55

No calculator Exam C · Q4

[Maximum mark: 4]

Match the expressions.

$c^{10} \div c^5$	c^{16}
$c^7 \times c^3$	c^5
$(c^8)^2$	c^2
$\frac{c^6}{c^4}$	c^{10}

56

No calculator Exam C · Q5

[Maximum mark: 2]

Simplify the following. $\frac{s^3 \times s^4}{s^2}$

54 *Exam C · Q3*

(a) $\boxed{w^8}$ (not w^{16}) **B1**

(b) $\boxed{a^3}$ (not a^2) **B1**

55 *Exam C · Q4*

$c^{10} \div c^5 = \boxed{c^5}$, $c^7 \times c^3 = \boxed{c^{10}}$, $(c^8)^2 = \boxed{c^{16}}$, $\dfrac{c^6}{c^4} = \boxed{c^2}$ **B4**

56 *Exam C · Q5*

$\dfrac{s^7}{s^2} = \boxed{s^5}$ **M1 A1**

57

No calculator **Exam C · Q7**

[Maximum mark: 2]

Simplify $2a^3c^3 \times 3a^2c$

58

No calculator **Exam C · Q8**

[Maximum mark: 2]

Simplify $\dfrac{10m^5n^4}{2m^2n}$

59

No calculator **Exam C · Q9**

[Maximum mark: 2]

Simplify $(2m^4)^3$

57 *Exam C · Q7*

$6a^{3+2}c^{3+1} = \boxed{6a^5c^4}$ **B2**

58 *Exam C · Q8*

$\boxed{5m^3n^3}$ **B2** (B1 for two of 5, m^3, n^3)

59 *Exam C · Q9*

$2^3 \times m^{12} = \boxed{8m^{12}}$ **B2** (not $6m^{12}$ or $8m^7$)

60

No calculator

MEP Pupil Text 1 · p.8 Q10

[Maximum mark: 24]

Simplify each of the following expressions.

(a) $a^3 \times a^2$ **(b)** $a^4 \times a^6$ **(c)** $x^2 \times x^7$

(d) $x^4 \div x^2$ **(e)** $y^3 \times y^0$ **(f)** $p^7 \div p^4$

(g) $q^6 \div q^3$ **(h)** $x^7 \times x$ **(i)** $b^4 \div b$

(j) $\dfrac{b^6}{b^0}$ **(k)** $\dfrac{c^7}{c^4}$ **(l)** $\dfrac{x^8}{x^3}$

(m) $\dfrac{y^3}{y}$ **(n)** $\dfrac{x^4}{x^4}$ **(o)** $x^2 \times x^3 \times x^3$

(p) $\dfrac{p^2 \times p^7}{p^5}$ **(q)** $\dfrac{x^{10}}{x^2 \times x^5}$ **(r)** $\dfrac{y^3 \times y^7}{y^2 \times y^4}$

(s) $\dfrac{x^2 \times x^3}{x^5}$ **(t)** $\dfrac{x^7 \times x}{x^3 \times x^4}$ **(u)** $\dfrac{x^8 \times x^4}{x^0}$

(v) $(x^2)^4$ **(w)** $(x^3)^5$ **(x)** $(x^2 \times x^7)^6$

60 *MEP Pupil Text 1 · p.8 Q10*

(a) a^5 **(b)** a^{10} **(c)** x^9 **(d)** x^2

(e) y^3 **(f)** p^3 **(g)** q^3 **(h)** x^8

(i) b^3 **(j)** b^6 **(k)** c^3 **(l)** x^5

(m) y^2 **(n)** $x^0 = 1$ **(o)** x^8 **(p)** p^4

(q) x^3 **(r)** y^4 **(s)** $x^0 = 1$ **(t)** $x^1 = x$

(u) x^{12} **(v)** x^8 **(w)** x^{15} **(x)** $(x^9)^6 = x^{54}$

(e) $y^0 = 1$, so $y^3 \times y^0 = y^3$. (h) $x = x^1$. (x) brackets first: $x^2 \times x^7 = x^9$, then $9 \times 6 = 54$.

61

No calculator PP p.1 · Q5

[Maximum mark: 1]

Write down the value of 9^0.

62

No calculator PP p.2 · Q4

[Maximum mark: 1]

Simplify. $a^2 \div a^6$

63

No calculator PP p.2 · Q6

[Maximum mark: 2]

Simplify.

(a) $y^3 \div y^5$ [1]

(b) $7x^0$ [1]

64

No calculator PP p.2 · Q3

[Maximum mark: 2]

Simplify. $(x^8y^7) \div (x^{-1}y^3)$

61 *PP p.1 · Q5*

$\boxed{1}$ **B1**

62 *PP p.2 · Q4*

$\boxed{a^{-4}}$ or $\dfrac{1}{a^4}$ **B1**

63 *PP p.2 · Q6*

(a) $\boxed{y^{-2}}$ or $\dfrac{1}{y^2}$ **B1**

(b) $7 \times x^0 = 7 \times 1 = \boxed{7}$ (not 1) **B1**

64 *PP p.2 · Q3*

$x^{8-(-1)}y^{7-3} = \boxed{x^9y^4}$ **B2** (B1 for x^9 or y^4)

65

No calculator PP p.10 · Q13

[Maximum mark: 3]

(a) Simplify $h^2 \times h^5$. [1]

(b) Simplify $\left(\frac{7}{x}\right)^{-3}$. [1]

(c) $a^8 \div a^p = a^2$. Find the value of p. [1]

66

No calculator PP p.11 · Q8

[Maximum mark: 1]

Find the value of **(a)** $\left(\frac{1}{2}\right)^{-3}$.

67

No calculator RR p.9 · Q20

[Maximum mark: 4]

Simplify.

(a) $6w^0$ [1]

(b) $5x^3 - 3x^3$ [1]

(c) $3y^6 \times 5y^{-2}$ [2]

68

No calculator RR p.9 · Q2

[Maximum mark: 1]

Work out. $2^{-4} \times 2^5$

65 *PP p.10 · Q13*

(a) $\boxed{h^7}$ **B1**

(b) $\left(\frac{x}{7}\right)^3 = \boxed{\frac{x^3}{343}}$ **B1**

(c) $8 - p = 2 \Rightarrow \boxed{p = 6}$ **B1**

66 *PP p.11 · Q8*

$\left(\frac{2}{1}\right)^3 = 2^3 = \boxed{8}$ **B1**

67 *RR p.9 · Q20*

(a) $6 \times 1 = \boxed{6}$ **B1**

(b) $\boxed{2x^3}$ (like terms — subtract the numbers, keep x^3) **B1**

(c) $\boxed{15y^4}$ **B2**

68 *RR p.9 · Q2*

$2^{-4+5} = 2^1 = \boxed{2}$ **B1**

69

No calculator RR p.5 · Q7

[Maximum mark: 2]

Simplify. $6uw^{-3} \times 4uw^{6}$

70

No calculator RR p.3 · Q14

[Maximum mark: 2]

(b) $5^{7} \div 5^{9} = p^{2}$. Find p.

71

No calculator RR p.10 · Q17

[Maximum mark: 4]

(a) $t^{x} \times t^{2} = t^{10}$. Find the value of x. [1]

(b) Simplify (i) $\left(\frac{4}{x}\right)^{-2}$ (ii) $a^{3}b^{7} \div a^{6}b^{2}$ [3]

72

No calculator RR p.8 · Q14

[Maximum mark: 1]

Work out. **(b)** $\left(\frac{1}{3}\right)^{-2}$

(part (a) $125^{\frac{2}{3}}$ is the same as PP p.12 Q17(a))

69 *RR p.5 · Q7*

$\boxed{24u^2w^3}$ **B2**

70 *RR p.3 · Q14*

$5^{-2} = p^2$ **M1**

$p^2 = \left(\frac{1}{5}\right)^2 \Rightarrow \boxed{p = \frac{1}{5}}$ (or 5^{-1}) **A1**

71 *RR p.10 · Q17*

(a) $x + 2 = 10 \Rightarrow \boxed{x = 8}$ **B1**

(b)(i) $\left(\frac{x}{4}\right)^2 = \boxed{\frac{x^2}{16}}$ **B1**

(ii) $a^{-3}b^5 = \boxed{\frac{b^5}{a^3}}$ **B2**

72 *RR p.8 · Q14*

$3^2 = \boxed{9}$ **B1**

73

No calculator **Exam A · Q8**

[Maximum mark: 5]

Work out the value of

(a) $(2^2)^3$ [1]

(b) $(\sqrt{3})^2$ [1]

(c) $\sqrt{2^4 \times 9}$ [2]

(d) 4^{-2} [1]

74

No calculator **Exam A · Q13**

[Maximum mark: 2]

Find the value of

(i) $36^{\frac{1}{2}}$ (ii) 3^{-2}

73 *Exam A · Q8*

(a) $2^6 = \boxed{64}$ **B1**

(b) $\boxed{3}$ **B1**

(c) $\sqrt{16 \times 9} = \sqrt{144} = \boxed{12}$ (or 4×3) **M1 A1**

(d) $\dfrac{1}{4^2} = \boxed{\dfrac{1}{16}}$ **B1**

74 *Exam A · Q13*

(i) $\sqrt{36} = \boxed{6}$ **B1**

(ii) $\dfrac{1}{3^2} = \boxed{\dfrac{1}{9}}$ **B1**

75

No calculator **Exam B · Q2**

[Maximum mark: 2]

Write down the value of

(a) 7^0 [1]

(b) 4^{-1} [1]

76

No calculator **Exam B · Q3**

[Maximum mark: 2]

(a) Simplify 2^0 [1]

(b) Simplify 5^{-1} [1]

77

No calculator **Exam B · Q4**

[Maximum mark: 2]

(a) Write down the value of 2^{-1} [1]

(b) Write down the value of $64^{\frac{1}{2}}$ [1]

75 *Exam B · Q2*

(a) $\boxed{1}$ **B1**

(b) $\boxed{\frac{1}{4}}$ (or 0.25) **B1**

76 *Exam B · Q3*

(a) $\boxed{1}$ **B1**

(b) $\boxed{\frac{1}{5}}$ (or 0.2) **B1**

77 *Exam B · Q4*

(a) $\boxed{\frac{1}{2}}$ **B1**

(b) $\sqrt{64} = \boxed{8}$ **B1**

78

No calculator **Exam B · Q5**

[Maximum mark: 3]

Write down the value of

(i) 5^0 (ii) 4^{-2} (iii) $100^{\frac{1}{2}}$

79

No calculator **Exam B · Q12**

[Maximum mark: 3]

(a) Work out $3^6 \div 3^{-7}$ [1]

(b) Write down the value of $36^{\frac{1}{2}}$ [1]

(c) $3^n = \frac{1}{9}$. Find the value of n. [1]

80

No calculator **Exam C · Q6**

[Maximum mark: 2]

(a) Simplify $w^3 \times w^{-5}$ [1]

(b) Simplify $a^4 \div a^{-2}$ [1]

78 *Exam B · Q5*

(i) $\boxed{1}$ **B1**

(ii) $\boxed{\frac{1}{16}}$ **B1**

(iii) $\boxed{10}$ **B1**

79 *Exam B · Q12*

(a) $3^{6-(-7)} = \boxed{3^{13}}$ **B1**

(b) $\boxed{6}$ **B1**

(c) $\frac{1}{9} = \frac{1}{3^2} = 3^{-2} \Rightarrow \boxed{n = -2}$ **B1**

80 *Exam C · Q6*

(a) $w^{3+(-5)} = \boxed{w^{-2}} = \frac{1}{w^2}$ **B1**

(b) $a^{4-(-2)} = \boxed{a^6}$ **B1**

81

No calculator PP p.5 · Q5

[Maximum mark: 1]

Find the value of **(b)** $64^{\frac{1}{3}}$.

(part (a) 64^0 is the same as Exam A Q20)

82

No calculator PP p.6 · Q12

[Maximum mark: 1]

Work out the value of $32^{\frac{2}{5}}$.

83

No calculator PP p.6 · Q5

[Maximum mark: 2]

Work out $4^{\frac{3}{2}}$. [1]

(b) $7^n = \sqrt[5]{7}$. Find the value of n. [1]

84

No calculator PP p.11 · Q8

[Maximum mark: 1]

Find the value of $125^{-\frac{1}{3}}$.

85

No calculator PP p.11 · Q11

[Maximum mark: 2]

Work out $4^{-\frac{3}{2}}$.

81 *PP p.5 · Q5*

$\sqrt[3]{64} = \boxed{4}$ **B1**

82 *PP p.6 · Q12*

$(\sqrt[5]{32})^2 = 2^2 = \boxed{4}$ **B1**

83 *PP p.6 · Q5*

(a) $(\sqrt{4})^3 = 2^3 = \boxed{8}$ **B1**

(b) $\sqrt[5]{7} = 7^{\frac{1}{5}} \Rightarrow \boxed{n = \frac{1}{5}}$ **B1**

84 *PP p.11 · Q8*

$\frac{1}{\sqrt[3]{125}} = \boxed{\frac{1}{5}}$ **B1**

85 *PP p.11 · Q11*

$\frac{1}{(\sqrt{4})^3} = \frac{1}{2^3} = \boxed{\frac{1}{8}}$ **M1 A1**

86

No calculator PP p.11 · Q19

[Maximum mark: 2]

Find the value of $27^{-\frac{2}{3}}$.

87

No calculator PP p.12 · Q17

[Maximum mark: 4]

Find the value of

(a) $125^{\frac{2}{3}}$ [2]

(b) $4^{-\frac{5}{2}}$ [2]

88

No calculator RR p.6 · Q10

[Maximum mark: 2]

Find the value of

(a) $(\sqrt{5})^8$ [1]

(b) $\left(\frac{1}{27}\right)^{-\frac{2}{3}}$ [1]

86 *PP p.11 · Q19*

$$\frac{1}{(\sqrt[3]{27})^2} = \frac{1}{3^2} = \boxed{\frac{1}{9}}$$ **M1 A1**

87 *PP p.12 · Q17*

(a) $(\sqrt[3]{125})^2 = 5^2 = \boxed{25}$ **M1 A1**

(b) $\frac{1}{(\sqrt{4})^5} = \frac{1}{2^5} = \boxed{\frac{1}{32}}$ **M1 A1**

88 *RR p.6 · Q10*

(a) $(5^{\frac{1}{2}})^8 = 5^4 = \boxed{625}$ **B1**

(b) $27^{\frac{2}{3}} = (\sqrt[3]{27})^2 = \boxed{9}$ **B1**

89

No calculator RR p.6 · Q16

[Maximum mark: 2]

(a) Find the value of

(i) $\left(\frac{1}{4}\right)^{0.5}$ [1]

(ii) $(-8)^{\frac{2}{3}}$ [1]

90

No calculator RR p.10 · Q5

[Maximum mark: 1]

Work out. $\left(\frac{125}{27}\right)^{-\frac{2}{3}}$

91

No calculator RR p.10 · Q11

[Maximum mark: 2]

Find the exact value of $8^{\frac{2}{3}} \times 49^{-\frac{1}{2}}$.

92

No calculator RR p.2 · Q10

[Maximum mark: 2]

Without using a calculator, show that $\left(\frac{49}{16}\right)^{-\frac{3}{2}} = \frac{64}{343}$.

Write down all the steps in your working.

89 *RR p.6 · Q16*

(i) $0.5 = \frac{1}{2}$, so $\sqrt{\frac{1}{4}} = \boxed{\frac{1}{2}}$ **B1**

(ii) $(\sqrt[3]{-8})^2 = (-2)^2 = \boxed{4}$ **B1**

90 *RR p.10 · Q5*

$\left(\frac{27}{125}\right)^{\frac{2}{3}} = \left(\frac{3}{5}\right)^2 = \boxed{\frac{9}{25}}$ **B1**

91 *RR p.10 · Q11*

$8^{\frac{2}{3}} = 4,\ 49^{-\frac{1}{2}} = \frac{1}{7}$ **M1**

$4 \times \frac{1}{7} = \boxed{\frac{4}{7}}$ **A1**

92 *RR p.2 · Q10*

$\left(\frac{16}{49}\right)^{\frac{3}{2}}$ **M1**

$= \left(\frac{4}{7}\right)^3 = \frac{64}{343}$ **A1**

93

Calculator RR p.3 · Q5

[Maximum mark: 2]

Write the following in order of size, **smallest** first.

$(1.5)^{\frac{2}{3}}$ $\left(\frac{2}{3}\right)^{1.5}$ $\left(\frac{2}{3}\right)^{-1.5}$ $\left(-\frac{2}{3}\right)^{\frac{2}{3}}$

$\ldots < \ldots < \ldots < \ldots$

94

No calculator Exam A · Q19

[Maximum mark: 4]

Work out

(i) 4^0 (ii) 4^{-2} (iii) $16^{\frac{3}{2}}$

95

No calculator Exam A · Q20

[Maximum mark: 4]

Find the value of

(i) 64^0 (ii) $64^{\frac{1}{2}}$ (iii) $64^{-\frac{2}{3}}$

93 *RR p.3 · Q5*

$1.5^{\frac{2}{3}} = 1.310,\ (\frac{2}{3})^{1.5} = 0.544,\ (\frac{2}{3})^{-1.5} = 1.837,\ (-\frac{2}{3})^{\frac{2}{3}} = \sqrt[3]{\frac{4}{9}} = 0.763$ **M1**

$$\boxed{\left(\frac{2}{3}\right)^{1.5} < \left(-\frac{2}{3}\right)^{\frac{2}{3}} < (1.5)^{\frac{2}{3}} < \left(\frac{2}{3}\right)^{-1.5}}$$ **A1**

TI-Nspire CX: Question 93

Type each one and press ctrl enter for a decimal: `1.5^(2/3)` → 1.31037; `(2/3)^1.5` → 0.544331; `(2/3)^(-1.5)` → 1.83712.

For $\left(-\frac{2}{3}\right)^{\frac{2}{3}}$ type `((-2/3)^2)^(1/3)` → 0.763143 (square first, then cube root) so the calculator does not give a complex or error answer. Use (-) for the negative sign.

94 *Exam A · Q19*

(i) $\boxed{1}$ **B1**

(ii) $\boxed{\frac{1}{16}}$ **B1**

(iii) $(\sqrt{16})^3 = 4^3 = \boxed{64}$ **M1 A1**

95 *Exam A · Q20*

(i) $\boxed{1}$ **B1**

(ii) $\boxed{8}$ **B1**

(iii) $\frac{1}{64^{\frac{2}{3}}} = \frac{1}{(\sqrt[3]{64})^2} = \frac{1}{4^2} = \boxed{\frac{1}{16}}$ **M1 A1**

96

No calculator **Exam B · Q6**

[Maximum mark: 4]

(a) Write down the value of (i) 9^0 (ii) $169^{\frac{1}{2}}$ [2]

(b) Work out $64^{\frac{2}{3}}$ [2]

97

No calculator **Exam B · Q7**

[Maximum mark: 3]

(a) Find the value of $36^{\frac{1}{2}}$ [1]

(b) Find the value of $8^{-\frac{2}{3}}$ [2]

98

No calculator **Exam B · Q9**

[Maximum mark: 2]

Write down the value of

(a) $25^{\frac{1}{2}}$ [1]

(b) 9^0 [1]

96 *Exam B · Q6*

(a)(i) $\boxed{1}$ **B1** (ii) $\boxed{13}$ **B1**

(b) $(\sqrt[3]{64})^2 = 4^2 = \boxed{16}$ **M1 A1**

97 *Exam B · Q7*

(a) $\boxed{6}$ **B1**

(b) $\dfrac{1}{(\sqrt[3]{8})^2} = \dfrac{1}{2^2} = \boxed{\dfrac{1}{4}}$ **M1 A1**

98 *Exam B · Q9*

(a) $\boxed{5}$ **B1**

(b) $\boxed{1}$ **B1**

99

No calculator **Exam B · Q10**

[Maximum mark: 5]

Evaluate

(i) 3^{-2} (ii) $36^{\frac{1}{2}}$ (iii) $27^{\frac{2}{3}}$ (iv) $\left(\frac{16}{81}\right)^{-\frac{3}{4}}$

100

No calculator **Exam B · Q11(b)**

[Maximum mark: 2]

$3 \times \sqrt{27} = 3^n$

Find the value of n.

(part (a) is the same as Exam A Q20 and is not repeated)

99 *Exam B · Q10*

(i) $\boxed{\frac{1}{9}}$ **B1**

(ii) $\boxed{6}$ **B1**

(iii) $(\sqrt[3]{27})^2 = 3^2 = \boxed{9}$ **B1**

(iv) $\left(\frac{81}{16}\right)^{\frac{3}{4}} = \left(\frac{\sqrt[4]{81}}{\sqrt[4]{16}}\right)^3 = \left(\frac{3}{2}\right)^3 = \boxed{\frac{27}{8}}$ **M1 A1**

Negative index on a fraction: **flip** the fraction and make the index positive.

100 *Exam B · Q11(b)*

$\sqrt{27} = (3^3)^{\frac{1}{2}} = 3^{\frac{3}{2}}$ **M1**

$3^1 \times 3^{\frac{3}{2}} = 3^{\frac{5}{2}} \Rightarrow \boxed{n = \frac{5}{2}}$ **A1**

101

No calculator **Exam B · Q16**

[Maximum mark: 6]

(a) Write down the value of $8^{\frac{1}{3}}$ [1]

$8\sqrt{8}$ can be written in the form 8^k.

(b) Find the value of k. [1]

$8\sqrt{8}$ can also be expressed in the form $m\sqrt{2}$ where m is a positive integer.

(c) Express $8\sqrt{8}$ in the form $m\sqrt{2}$. [2]

(d) Rationalise the denominator of $\frac{1}{8\sqrt{8}}$. Give your answer in the form $\frac{\sqrt{2}}{p}$ where p is a positive integer. [2]

102

No calculator **MEP Pupil Text 1 · p.14 Q1**

[Maximum mark: 15]

Find as fractions that do not involve indices, *without* using a calculator:

(a) 4^{-2} **(b)** 2^{-3} **(c)** 6^{-1}

(d) 7^{-1} **(e)** $9^{\frac{1}{2}}$ **(f)** $64^{\frac{1}{2}}$

(g) $16^{\frac{1}{4}}$ **(h)** $27^{\frac{1}{3}}$ **(i)** $1^{\frac{1}{3}}$

(j) 5^{-2} **(k)** $16^{\frac{3}{4}}$ **(l)** $4^{\frac{5}{2}}$

(m) $9^{\frac{7}{2}}$ **(n)** $25^{\frac{3}{2}}$ **(o)** $8^{-\frac{1}{3}}$

101 *Exam B · Q16*

(a) $\boxed{2}$ **B1**

(b) $8^1 \times 8^{\frac{1}{2}} = 8^{\frac{3}{2}}$, $\boxed{k = \frac{3}{2}}$ **B1**

(c) $\sqrt{8} = \sqrt{4 \times 2} = 2\sqrt{2}$, so $8 \times 2\sqrt{2} = \boxed{16\sqrt{2}}$ **M1 A1**

(d) $\frac{1}{16\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{16 \times 2} = \boxed{\frac{\sqrt{2}}{32}}$ **M1 A1**

102 *MEP Pupil Text 1 · p.14 Q1*

(a) $4^{-2} = \frac{1}{16}$ **(b)** $2^{-3} = \frac{1}{8}$ **(c)** $6^{-1} = \frac{1}{6}$

(d) $7^{-1} = \frac{1}{7}$ **(e)** $9^{\frac{1}{2}} = 3$ **(f)** $64^{\frac{1}{2}} = 8$

(g) $16^{\frac{1}{4}} = 2$ **(h)** $27^{\frac{1}{3}} = 3$ **(i)** $1^{\frac{1}{3}} = 1$

(j) $5^{-2} = \frac{1}{25}$ **(k)** $16^{\frac{3}{4}} = 2^3 = 8$ **(l)** $4^{\frac{5}{2}} = 2^5 = 32$

(m) $9^{\frac{7}{2}} = 3^7 = 2187$ **(n)** $25^{\frac{3}{2}} = 5^3 = 125$ **(o)** $8^{-\frac{1}{3}} = \frac{1}{2}$

Fractional index: **root first, then power**, e.g. $16^{\frac{3}{4}} = (\sqrt[4]{16})^3 = 2^3$.

103

No calculator

MEP Pupil Text 1 · p.14 Q2

[Maximum mark: 18]

Complete the missing numbers, *without* using a calculator.

(a) $3^{?} = \frac{1}{81}$ **(b)** $2^{?} = \frac{1}{2}$ **(c)** $5^{?} = \frac{1}{125}$

(d) $36^{?} = 6$ **(e)** $36^{?} = \frac{1}{6}$ **(f)** $7^{?} = 49$

(g) $7^{?} = 343$ **(h)** $17^{?} = \frac{1}{17}$ **(i)** $125^{?} = 5$

(j) $\frac{1}{2} = 2^{?}$ **(k)** $\frac{1}{4} = 2^{?}$ **(l)** $\frac{1}{100} = 10^{?}$

(m) $\frac{1}{a^3} = a^{?}$ **(n)** $\sqrt{m} = m^{?}$ **(o)** $\frac{1}{p^2} = p^{?}$

(p) $\sqrt[3]{q} = q^{?}$ **(q)** $\sqrt[3]{q^2} = q^{?}$ **(r)** $\sqrt[5]{q^2} = q^{?}$

104

Calculator

MEP Pupil Text 1 · p.15 Q3

[Maximum mark: 12]

Use a calculator to find:

(a) 8^{-1} **(b)** 20^{-1} **(c)** $\left(\frac{1}{2}\right)^{-1}$ **(d)** $\left(\frac{1}{4}\right)^{-1}$

(e) 15^{-2} **(f)** 20^{-3} **(g)** $81^{\frac{3}{2}}$ **(h)** $243^{\frac{3}{5}}$

(i) $16^{-\frac{1}{4}}$ **(j)** $144^{\frac{3}{2}}$ **(k)** $169^{\frac{7}{2}}$ **(l)** $121^{\frac{3}{2}}$

103 *MEP Pupil Text 1 · p.14 Q2*

(a) -4 **(b)** -1 **(c)** -3 **(d)** $\frac{1}{2}$ **(e)** $-\frac{1}{2}$ **(f)** 2

(g) 3 **(h)** -1 **(i)** $\frac{1}{3}$ **(j)** -1 **(k)** -2 **(l)** -2

(m) -3 **(n)** $\frac{1}{2}$ **(o)** -2 **(p)** $\frac{1}{3}$ **(q)** $\frac{2}{3}$ **(r)** $\frac{2}{5}$

$\frac{1}{a^n} = a^{-n}$; $\sqrt[n]{a} = a^{\frac{1}{n}}$; $\sqrt[n]{a^m} = a^{\frac{m}{n}}$.

104 *MEP Pupil Text 1 · p.15 Q3*

(a) $8^{-1} = 0.125$ **(b)** $20^{-1} = 0.05$ **(c)** $(\frac{1}{2})^{-1} = 2$

(d) $(\frac{1}{4})^{-1} = 4$ **(e)** $15^{-2} = \frac{1}{225} = 0.00\dot{4}$ **(f)** $20^{-3} = \frac{1}{8000} = 0.000125$

(g) $81^{\frac{3}{2}} = 729$ **(h)** $243^{\frac{3}{5}} = 27$ **(i)** $16^{-\frac{1}{4}} = 0.5$

(j) $144^{\frac{3}{2}} = 1728$ **(k)** $169^{\frac{7}{2}} = 62\,748\,517$ **(l)** $121^{\frac{3}{2}} = 1331$

TI-Nspire CX: Question 104

home → **1: New** → **Add Calculator**. Type the base, press ^ and type the index, then enter.

- Negative index: use the (-) key (not the minus key), e.g. `8^(-1)`.
- Fractional index: put the fraction in brackets, e.g. `81^(3/2)`, or use the fraction template ctrl ÷.
- Answer shown as a fraction? Press ctrl enter for a decimal (≈).
- Roots: ctrl x^2 gives $\sqrt{}$; ctrl ^ gives the nth-root template $\sqrt[n]{}$.

e.g. `169^(7/2)` enter → 62748517; `15^(-2)` ctrl enter → 0.004444.

105

Calculator **Practice**

[Maximum mark: 4]

Use your calculator to evaluate, giving answers exactly or to 3 significant figures.

(a) 1.5^{-3} **(b)** 0.8^{-2} **(c)** $7^{\frac{2}{3}}$ **(d)** $\left(\frac{2}{5}\right)^{-2}$

105 *Practice*

(a) $\dfrac{1}{3.375} = \boxed{0.296}$ **A1**

(b) $\dfrac{1}{0.64} = \boxed{1.5625}$ **A1**

(c) $\boxed{3.66}$ $(3.6593\ldots)$ **A1**

(d) $\left(\dfrac{5}{2}\right)^2 = \boxed{6.25}$ **A1**

TI-Nspire CX: Question 105

`1.5^(-3)` enter $\to 0.296296$; `0.8^(-2)` $\to 1.5625$; `7^(2/3)` ctrl enter $\to 3.65931$; `(2/5)^(-2)` $\to \frac{25}{4}$.

Use the (-) key for the negative sign, and brackets round fractional indices: `7^2/3` would give $49/3$ — wrong!

106

No calculator PP p.7 · Q15

[Maximum mark: 2]

Simplify $(3125w^{3125})^{\frac{1}{5}}$.

107

No calculator PP p.7 · Q18

[Maximum mark: 2]

Simplify $(343x^{9})^{\frac{2}{3}}$.

108

No calculator PP p.7 · Q17

[Maximum mark: 2]

Simplify $(81x^{12})^{\frac{3}{4}}$.

109

No calculator PP p.7 · Q22

[Maximum mark: 2]

Simplify $(216y^{216})^{\frac{2}{3}}$.

110

No calculator PP p.7 · Q14

[Maximum mark: 2]

Simplify $(25x^{16})^{\frac{3}{2}}$.

106 *PP p.7 · Q15*

$\sqrt[5]{3125} \times w^{3125 \div 5} = \boxed{5w^{625}}$ **B2**

107 *PP p.7 · Q18*

$(\sqrt[3]{343})^2 x^{9 \times \frac{2}{3}} = 7^2 x^6 = \boxed{49x^6}$ **B2**

108 *PP p.7 · Q17*

$(\sqrt[4]{81})^3 x^{12 \times \frac{3}{4}} = 3^3 x^9 = \boxed{27x^9}$ **B2**

109 *PP p.7 · Q22*

$(\sqrt[3]{216})^2 y^{216 \times \frac{2}{3}} = 6^2 y^{144} = \boxed{36y^{144}}$ **B2**

110 *PP p.7 · Q14*

$(\sqrt{25})^3 x^{16 \times \frac{3}{2}} = 5^3 x^{24} = \boxed{125x^{24}}$ **B2**

111

No calculator PP p.8 · Q17

[Maximum mark: 2]

Simplify $(64y^{27})^{\frac{2}{3}}$.

112

No calculator PP p.8 · Q21

[Maximum mark: 2]

Simplify $(243y^{10})^{\frac{3}{5}}$.

113

No calculator PP p.8 · Q21

[Maximum mark: 4]

Simplify.

(a) $(5x^4)^3$ [2]

(b) $(256x^{256})^{\frac{3}{8}}$ [2]

114

No calculator PP p.8 · Q7

[Maximum mark: 4]

Simplify.

(a) $\frac{15w^{15}}{3w^3}$ [2]

(b) $(125y^6)^{\frac{2}{3}}$ [2]

111 *PP p.8 · Q17*

$(\sqrt[3]{64})^2 y^{18} = 4^2 y^{18} = \boxed{16y^{18}}$ **B2**

112 *PP p.8 · Q21*

$(\sqrt[5]{243})^3 y^6 = 3^3 y^6 = \boxed{27y^6}$ **B2**

113 *PP p.8 · Q21*

(a) $\boxed{125x^{12}}$ **B2**

(b) $(\sqrt[8]{256})^3 x^{256\times\frac{3}{8}} = 2^3 x^{96} = \boxed{8x^{96}}$ **B2**

114 *PP p.8 · Q7*

(a) $\boxed{5w^{12}}$ **B2**

(b) $(\sqrt[3]{125})^2 y^4 = \boxed{25y^4}$ **B2**

115

No calculator PP p.9 · Q23

[Maximum mark: 2]

(a) Simplify $\left(\frac{x^2}{4}\right)^{\frac{3}{2}}$.

116

No calculator PP p.13 · Q20

[Maximum mark: 3]

Simplify fully.

(a) $(81x^{16})^{\frac{3}{4}}$ [2]

(b) $\left(\frac{1}{y^2}\right)^{-\frac{1}{2}}$ [1]

117

No calculator PP p.13 · Q2

[Maximum mark: 6]

(a) Simplify fully.

(i) $p^3 \times p^{11}$ [1]

(ii) $\frac{18m^6}{3m^2}$ [2]

(iii) $\left(\frac{27x^9y^{27}}{64}\right)^{-\frac{1}{3}}$ [3]

115 *PP p.9 · Q23*

$$\frac{x^{2\times\frac{3}{2}}}{(\sqrt{4})^3} = \boxed{\frac{x^3}{8}}$$ **B2**

116 *PP p.13 · Q20*

(a) $3^3x^{12} = \boxed{27x^{12}}$ **B2**

(b) $(y^2)^{\frac{1}{2}} = \boxed{y}$ **B1**

117 *PP p.13 · Q2*

(i) $\boxed{p^{14}}$ **B1**

(ii) $\boxed{6m^4}$ **B2**

(iii) $\left(\frac{64}{27x^9y^{27}}\right)^{\frac{1}{3}}$ **M1** $= \frac{4}{3x^3y^9}$, $\boxed{\frac{4}{3x^3y^9}}$ **A2**

118

No calculator PP p.14 · Q9

[Maximum mark: 7]

(a) Simplify.

(i) $(3x^2y^4)^3$ [2]

(ii) $\left(\frac{16}{x^{16}y^8}\right)^{-\frac{3}{2}}$ [3]

(b) $2^{12} \div 2^{\frac{k}{2}} = 32$. Find the value of k. [2]

119

No calculator RR p.1 · Q14

[Maximum mark: 4]

Simplify the following.

(a) $(4pq^2)^3$ [2]

(b) $(16x^8)^{-\frac{1}{4}}$ [2]

120

No calculator RR p.1 · Q5

[Maximum mark: 2]

Simplify the expression. $\left(a^{\frac{1}{2}} - b^{\frac{1}{2}}\right)\left(a^{\frac{1}{2}} + b^{\frac{1}{2}}\right)$

118 *PP p.14 · Q9*

(a)(i) $\boxed{27x^6y^{12}}$ **B2**

(ii) $\left(\dfrac{x^{16}y^8}{16}\right)^{\frac{3}{2}}$ **M1** $= \dfrac{x^{24}y^{12}}{(\sqrt{16})^3} = \boxed{\dfrac{x^{24}y^{12}}{64}}$ **A2**

(b) $2^{12-\frac{k}{2}} = 2^5 \Rightarrow 12 - \dfrac{k}{2} = 5$ **M1** $\Rightarrow \boxed{k = 14}$ **A1**

119 *RR p.1 · Q14*

(a) $\boxed{64p^3q^6}$ **B2**

(b) $\dfrac{1}{16^{\frac{1}{4}}x^2} = \boxed{\dfrac{1}{2x^2}}$ **B2**

120 *RR p.1 · Q5*

Difference of two squares: $(a^{\frac{1}{2}})^2 - (b^{\frac{1}{2}})^2$ **M1**

$= \boxed{a-b}$ **A1**

121

No calculator RR p.2 · Q11

[Maximum mark: 2]

Simplify $(256w^{256})^{\frac{1}{4}}$.

122

No calculator RR p.2 · Q11

[Maximum mark: 2]

Write $(27x^{12})^{\frac{1}{3}}$ in its simplest form.

123

No calculator RR p.3 · Q13

[Maximum mark: 3]

(a) $3^x = \sqrt[4]{3^5}$. Find the value of x. [1]

(b) Simplify $(32y^{15})^{\frac{2}{5}}$. [2]

124

No calculator RR p.3 · Q14

[Maximum mark: 2]

(a) Simplify $(64q^{-2})^{\frac{1}{2}}$.

121 *RR p.2 · Q11*

$\boxed{4w^{64}}$ **B2**

122 *RR p.2 · Q11*

$\boxed{3x^4}$ **B2**

123 *RR p.3 · Q13*

(a) $3^{\frac{5}{4}} \Rightarrow \boxed{x = \frac{5}{4}}$ **B1**

(b) $(\sqrt[5]{32})^2 y^6 = \boxed{4y^6}$ **B2**

124 *RR p.3 · Q14*

$8q^{-1} = \boxed{\frac{8}{q}}$ **B2**

125

No calculator **RR p.4 · Q17**

[Maximum mark: 3]

(a) Simplify $(3125t^{125})^{\frac{1}{5}}$. [2]

(c) Find the value of w when $x^{72} \div x^{w} = x^{8}$. [1]

(part (b) is the same as Exam B Q12(c))

126

No calculator **RR p.5 · Q11**

[Maximum mark: 2]

(b) Simplify $\left(\frac{x^6}{27}\right)^{\frac{1}{3}}$.

(part (a) is the same as PP p.1 Q5)

127

No calculator **RR p.6 · Q13**

[Maximum mark: 4]

Simplify.

(a) $12x^{12} \div 3x^{3}$ [2]

(b) $(256y^{256})^{\frac{1}{8}}$ [2]

125 *RR p.4 · Q17*

(a) $\boxed{5t^{25}}$ **B2**

(c) $72 - w = 8 \Rightarrow \boxed{w = 64}$ **B1**

126 *RR p.5 · Q11*

$\boxed{\dfrac{x^2}{3}}$ **B2**

127 *RR p.6 · Q13*

(a) $\boxed{4x^9}$ **B2**

(b) $\boxed{2y^{32}}$ **B2**

128

No calculator RR p.6 · Q12

[Maximum mark: 2]

Simplify $(16p^{16})^{\frac{1}{4}}$.

129

No calculator RR p.7 · Q17

[Maximum mark: 3]

Simplify. $\left(\frac{x^{64}}{16y^{16}}\right)^{\frac{1}{4}}$

130

No calculator RR p.7 · Q6

[Maximum mark: 2]

Simplify $\left(\frac{1}{2}x^{\frac{2}{3}}\right)^{3}$.

131

No calculator RR p.7 · Q7

[Maximum mark: 2]

Simplify $(32x^{10})^{\frac{3}{5}}$.

132

No calculator RR p.7 · Q10

[Maximum mark: 2]

Simplify $(36x^{16})^{\frac{1}{2}}$.

128 *RR p.6 · Q12*

$\boxed{2p^4}$ **B2**

129 *RR p.7 · Q17*

$\dfrac{x^{16}}{16^{\frac{1}{4}}y^4}$ **M1** $= \boxed{\dfrac{x^{16}}{2y^4}}$ **A2**

130 *RR p.7 · Q6*

$\dfrac{1}{2^3}x^{\frac{2}{3}\times 3} = \boxed{\dfrac{x^2}{8}}$ **B2**

131 *RR p.7 · Q7*

$(\sqrt[5]{32})^3x^6 = \boxed{8x^6}$ **B2**

132 *RR p.7 · Q10*

$\boxed{6x^8}$ **B2**

133

No calculator RR p.8 · Q25

[Maximum mark: 4]

(a) Simplify $(16x^{16})^{\frac{3}{4}}$. [2]

(b) $2p^{\frac{3}{2}} = 54$. Find the value of p. [2]

134

No calculator RR p.8 · Q8

[Maximum mark: 2]

Simplify $\left(\frac{8}{a^{12}}\right)^{\frac{1}{3}}$.

135

No calculator RR p.9 · Q27

[Maximum mark: 4]

Simplify.

(a) $81^{\frac{3}{4}}$ [1]

(b) $x^{\frac{2}{3}} \div x^{-\frac{4}{3}}$ [1]

(c) $\left(\frac{8}{y^6}\right)^{-\frac{1}{3}}$ [2]

133 *RR p.8 · Q25*

(a) $2^3x^{12} = \boxed{8x^{12}}$ **B2**

(b) $p^{\frac{3}{2}} = 27$ **M1**$\Rightarrow p = 27^{\frac{2}{3}} = \boxed{9}$ **A1**

134 *RR p.8 · Q8*

$\boxed{\dfrac{2}{a^4}}$ **B2**

135 *RR p.9 · Q27*

(a) $3^3 = \boxed{27}$ **B1**

(b) $x^{\frac{2}{3}+\frac{4}{3}} = \boxed{x^2}$ **B1**

(c) $\left(\dfrac{y^6}{8}\right)^{\frac{1}{3}} = \boxed{\dfrac{y^2}{2}}$ **B2**

136

No calculator RR p.11 · Q27

[Maximum mark: 5]

(a) Simplify $(27x^6)^{\frac{1}{3}}$. [2]

(b) Find the value of $(64x^4)^{0.5} \times 4x^{-2}$. [3]

137

No calculator Exam B · Q13

[Maximum mark: 4]

Simplify

(i) $(3x^2y)^3$ [2]

(ii) $(2t^{-3})^{-2}$ [2]

138

No calculator Exam C · Q10

[Maximum mark: 3]

Simplify $\dfrac{a^{\frac{1}{5}} \times a^{\frac{2}{3}}}{a^{\frac{3}{5}}}$

136 *RR p.11 · Q27*

(a) $\boxed{3x^2}$ **B2**

(b) $8x^2 \times 4x^{-2}$ **M1 M1** $= 32x^0 = \boxed{32}$ **A1**

137 *Exam B · Q13*

(i) $3^3x^6y^3 = \boxed{27x^6y^3}$ **B2**

(ii) $2^{-2}t^6 = \boxed{\frac{t^6}{4}}$ **B2**

(B1 for $\frac{1}{4}$ or t^6 seen)

138 *Exam C · Q10*

Numerator: $\frac{1}{5} + \frac{2}{3} = \frac{3}{15} + \frac{10}{15} = \frac{13}{15}$ **M1**

$\frac{13}{15} - \frac{3}{5} = \frac{13}{15} - \frac{9}{15} = \frac{4}{15}$ **M1**

$\boxed{a^{\frac{4}{15}}}$ **A1**

139

No calculator

Exam C · Q11

[Maximum mark: 3]

Simplify fully $\sqrt{\dfrac{50\pi a^5}{2\pi a^3}}$

140

No calculator

MEP Pupil Text 1 · p.15 Q4

[Maximum mark: 15]

Simplify the following expressions, so that they contain no negative indices.

(a) $a^6 \times a^{-7}$ **(b)** $\dfrac{a^7}{a^{-3}}$ **(c)** $\dfrac{a^{-5}}{a^{-9}}$

(d) $a^{-4} \times a^{-2}$ **(e)** $(a^2)^{-1}$ **(f)** $(a^2)^{-3}$

(g) $(a^{-2})^{-4}$ **(h)** $(a^{\frac{1}{2}})^5$ **(i)** $(a^3)^{-\frac{1}{2}}$

(j) $(a^6)^{\frac{1}{3}}$ **(k)** $(a^9)^{-\frac{1}{3}}$ **(l)** $(a^{-12})^{-\frac{1}{4}}$

(m) $\left(\dfrac{a}{b}\right)^2$ **(n)** $(a^2 \times b^{-4})^3$ **(o)** $(a^3 b^{\frac{1}{2}})^4$

139 *Exam C · Q11*

$\dfrac{50\pi a^5}{2\pi a^3} = 25a^2$ **M1 A1**

$\sqrt{25a^2} = \boxed{5a}$ **A1**

140 *MEP Pupil Text 1 · p.15 Q4*

(a) $a^6 \times a^{-7} = a^{-1} = \dfrac{1}{a}$ **(b)** $\dfrac{a^7}{a^{-3}} = a^{10}$ **(c)** $\dfrac{a^{-5}}{a^{-9}} = a^4$

(d) $a^{-4} \times a^{-2} = \dfrac{1}{a^6}$ **(e)** $(a^2)^{-1} = \dfrac{1}{a^2}$ **(f)** $(a^2)^{-3} = \dfrac{1}{a^6}$

(g) $(a^{-2})^{-4} = a^8$ **(h)** $(a^{\frac{1}{2}})^5 = a^{\frac{5}{2}}$ **(i)** $(a^3)^{-\frac{1}{2}} = \dfrac{1}{a^{\frac{3}{2}}}$

(j) $(a^6)^{\frac{1}{3}} = a^2$ **(k)** $(a^9)^{-\frac{1}{3}} = \dfrac{1}{a^3}$ **(l)** $(a^{-12})^{-\frac{1}{4}} = a^3$

(m) $\left(\dfrac{a}{b}\right)^2 = \dfrac{a^2}{b^2}$ **(n)** $(a^2 \times b^{-4})^3 = \dfrac{a^6}{b^{12}}$ **(o)** $(a^3 b^{\frac{1}{2}})^4 = a^{12}b^2$

(b) $7-(-3)=10$; (c) $-5-(-9)=4$; (g) $-2\times -4=8$; (l) $-12\times -\dfrac{1}{4}=3$.

141

 No calculator **Practice**

[Maximum mark: 5]

Simplify, giving answers without negative indices.

(a) $(8x^6)^{\frac{2}{3}}$ **(b)** $(27a^{-3})^{-\frac{1}{3}}$ **(c)** $(4x^{-2})^{-\frac{1}{2}}$

(d) $\dfrac{x^{\frac{1}{2}} \times x^{\frac{3}{2}}}{x^{-1}}$ **(e)** $(9a^4b^{-2})^{\frac{1}{2}}$

141 *Practice*

(a) $(\sqrt[3]{8})^2 x^{6\times\frac{2}{3}} = \boxed{4x^4}$ **A1**

(b) $27^{-\frac{1}{3}} a^1 = \boxed{\frac{a}{3}}$ **A1**

(c) $4^{-\frac{1}{2}} x^1 = \boxed{\frac{x}{2}}$ **A1**

(d) $\frac{x^2}{x^{-1}} = \boxed{x^3}$ **A1**

(e) $3a^2 b^{-1} = \boxed{\frac{3a^2}{b}}$ **A1**

142

No calculator PP p.1 · Q11

[Maximum mark: 1]

Find the value of p when $6^p \times 6^4 = 6^{28}$.

143

No calculator PP p.2 · Q10

[Maximum mark: 1]

$5^w \div 5^{13} = 25$. Find the value of w.

144

No calculator PP p.10 · Q15

[Maximum mark: 1]

$4^x = \frac{1}{64}$. Find the value of x.

145

No calculator PP p.9 · Q26

[Maximum mark: 2]

$\sqrt[3]{y^2} = \sqrt[6]{x}$ and $y = \sqrt[n]{x}$.

Find the value of n.

142 *PP p.1 · Q11*

$p+4=28$

$\boxed{p=24}$ **A1**

143 *PP p.2 · Q10*

$5^{w-13}=5^2 \Rightarrow w-13=2$

$\boxed{w=15}$ **A1**

144 *PP p.10 · Q15*

$\frac{1}{64}=\frac{1}{4^3}=4^{-3}$

$\boxed{x=-3}$ **A1**

145 *PP p.9 · Q26*

$y^{\frac{2}{3}}=x^{\frac{1}{6}}$ **M1**

raise both sides to the power $\frac{3}{2}$: $y=x^{\frac{1}{6}\times\frac{3}{2}}=x^{\frac{1}{4}}$, so $\boxed{n=4}$ **A1**

146

No calculator PP p.9 · Q21

[Maximum mark: 3]

(a) $\sqrt[5]{3} = 3^h$. Write down the value of h. [1]

(b) Simplify $(4x^3)^3$. [2]

147

No calculator PP p.15 · Q13

[Maximum mark: 6]

(a) $3^{3p} \times 3^{2p} = 729$. Find the value of p. [2]

(b) Simplify. $(32x^{10})^{\frac{1}{5}}$ [2]

(c) $(243y^{20})^{\frac{2}{5}}$ [2]

148

No calculator PP p.15 · Q6

[Maximum mark: 3]

Find the value of p when $2^6 \div 4^p = 2^7$.

146 *PP p.9 · Q21*

(a) $\boxed{h = \frac{1}{5}}$ **B1**

(b) $\boxed{64x^9}$ **B2**

147 *PP p.15 · Q13*

(a) $3^{5p} = 3^6$ **M1**$\Rightarrow \boxed{p = \frac{6}{5}}$ **A1**

(b) $\boxed{2x^2}$ **B2**

(c) $(\sqrt[5]{243})^2 y^8 = \boxed{9y^8}$ **B2**

148 *PP p.15 · Q6*

$4^p = (2^2)^p = 2^{2p}$ **M1**

$2^{6-2p} = 2^7 \Rightarrow 6 - 2p = 7$ **M1**

$\boxed{p = -\frac{1}{2}}$ **A1**

149

No calculator

PP p.16 · Q11

[Maximum mark: 5]

(a) Simplify fully. $(4ab^5)^4$ [2]

(b) $2p^{\frac{1}{3}} = 6$. Find the value of p. [1]

(c) $81^2 \div 3^t = 9$. Find the value of t. [2]

150

No calculator

PP p.16 · Q17

[Maximum mark: 2]

$3^x = 27^{x+2}$. Find the value of x.

151

No calculator

PP p.17 · Q16

[Maximum mark: 2]

$8^{\frac{4}{3}} = 32^x$. Find the value of x.

152

No calculator

PP p.17 · Q13

[Maximum mark: 2]

(a) Write 243×27^{2n} as a single power of 3 in terms of n.

149 *PP p.16 · Q11*

(a) $\boxed{256a^4b^{20}}$ **B2**

(b) $p^{\frac{1}{3}} = 3 \Rightarrow p = 3^3 = \boxed{27}$ **B1**

(c) $(3^4)^2 \div 3^t = 3^2 \Rightarrow 8 - t = 2$ **M1**$\Rightarrow \boxed{t = 6}$ **A1**

150 *PP p.16 · Q17*

$3^x = 3^{3(x+2)} \Rightarrow x = 3x + 6$ **M1**

$\boxed{x = -3}$ **A1**

151 *PP p.17 · Q16*

$8^{\frac{4}{3}} = (\sqrt[3]{8})^4 = 2^4$ and $32^x = 2^{5x}$ **M1**

$5x = 4 \Rightarrow \boxed{x = \frac{4}{5}}$ **A1**

152 *PP p.17 · Q13*

$3^5 \times (3^3)^{2n} = 3^5 \times 3^{6n}$ **M1**

$= \boxed{3^{5+6n}}$ **A1**

153

No calculator PP p.17 · Q18

[Maximum mark: 3]

(a) Simplify. $(4xy^2)^3$ [2]

(b) $25 = 125^k$. Find the value of k. [1]

154

No calculator PP p.18 · Q25

[Maximum mark: 3]

$m^{-\frac{1}{4}} = 27m^{-1}$. Find the value of m.

155

No calculator PP p.18 · Q(d)

[Maximum mark: 2]

$5^n + 5^n + 5^n + 5^n + 5^n = 5^m$.

Find an expression for m in terms of n.

153 *PP p.17 · Q18*

(a) $\boxed{64x^3y^6}$ **B2**

(b) $5^2 = 5^{3k} \Rightarrow \boxed{k = \frac{2}{3}}$ **B1**

154 *PP p.18 · Q25*

Multiply both sides by m: $m^{-\frac{1}{4}+1} = m^{\frac{3}{4}} = 27$ **M1**

$m = 27^{\frac{4}{3}} = (\sqrt[3]{27})^4 = 3^4$ **M1**

$\boxed{m = 81}$ **A1**

155 *PP p.18 · Q(d)*

$5 \times 5^n = 5^1 \times 5^n = 5^{n+1}$ **M1**

$\boxed{m = n+1}$ **A1**

156

No calculator PP p.19 · Q22

[Maximum mark: 5]

(a) Simplify. $\dfrac{x^{\frac{2}{3}}}{x^{\frac{8}{3}}}$ [1]

(b) $16 = 64^{k}$. Find the value of k. [1]

(c) Solve. $3^{3x} \times \left(\dfrac{1}{9}\right)^{4-3x} = 3$ [3]

157

No calculator PP p.20 · Q21

[Maximum mark: 7]

$\dfrac{16^{5m}}{4} = 64^{2n}$. Find m in terms of n. [3]

(b) $16^{x} \times \left(\dfrac{1}{2}\right)^{x} = 4^{x+3}$. Find the value of x. [4]

158

No calculator PP p.21 · Q(e)

[Maximum mark: 4]

$3^{2x-1} = \dfrac{1}{9^{x}} \times 3^{2y-x}$.

Find an expression for y in terms of x.

156 *PP p.19 · Q22*

(a) $x^{\frac{2}{3}-\frac{8}{3}} = x^{-2} = \boxed{\dfrac{1}{x^2}}$ **B1**

(b) $4^2 = 4^{3k} \Rightarrow \boxed{k = \dfrac{2}{3}}$ **B1**

(c) $\dfrac{1}{9} = 3^{-2}$: $3^{3x} \times 3^{-8+6x} = 3^1$ **M1**

$9x - 8 = 1$ **M1**$\Rightarrow \boxed{x = 1}$ **A1**

157 *PP p.20 · Q21*

(a) base 4: $\dfrac{4^{10m}}{4^1} = 4^{6n}$ **M1** $\Rightarrow 10m - 1 = 6n$ **M1**

$\boxed{m = \dfrac{6n+1}{10}}$ **A1**

(b) base 2: $2^{4x} \times 2^{-x} = 2^{2(x+3)}$ **M1 M1**

$3x = 2x + 6$ **M1**$\Rightarrow \boxed{x = 6}$ **A1**

158 *PP p.21 · Q(e)*

$\dfrac{1}{9^x} = 3^{-2x}$ **M1**

$3^{2x-1} = 3^{-2x+2y-x} = 3^{2y-3x}$ **M1**

$2x - 1 = 2y - 3x$ **M1**$\Rightarrow \boxed{y = \dfrac{5x-1}{2}}$ **A1**

159

No calculator RR p.1 · Q8

[Maximum mark: 2]

Find r when $5^{\frac{r}{3}} = 125$.

160

No calculator RR p.2 · Q14

[Maximum mark: 4]

(a) $\left(\frac{3}{8}\right)^{\frac{3}{8}} \times \left(\frac{3}{8}\right)^{\frac{1}{8}} = p^q$. Find the value of p and the value of q. [2]

(b) $5^{-3} + 5^{-4} = k \times 5^{-4}$. Find the value of k. [2]

161

No calculator RR p.5 · Q3

[Maximum mark: 1]

$81^x = 3$. Find the value of x.

162

No calculator RR p.8 · Q6

[Maximum mark: 1]

(a) $2^r = \frac{1}{16}$. Find the value of r.

(part (b) is the same as PP p.9 Q21(a))

159 *RR p.1 · Q8*

$5^{\frac{r}{3}} = 5^3$ **M1** $\Rightarrow \frac{r}{3} = 3 \Rightarrow \boxed{r = 9}$ **A1**

160 *RR p.2 · Q14*

(a) $\left(\frac{3}{8}\right)^{\frac{4}{8}} = \left(\frac{3}{8}\right)^{\frac{1}{2}}$: $\boxed{p = \frac{3}{8},\ q = \frac{1}{2}}$ **B2**

(b) $5^{-3} = 5 \times 5^{-4}$, so $5 \times 5^{-4} + 1 \times 5^{-4} = 6 \times 5^{-4}$ **M1** $\boxed{k = 6}$ **A1**

161 *RR p.5 · Q3*

$81^{\frac{1}{4}} = \sqrt[4]{81} = 3$

$\boxed{x = \frac{1}{4}}$ **A1**

162 *RR p.8 · Q6*

$\frac{1}{16} = 2^{-4} \Rightarrow \boxed{r = -4}$ **B1**

163

No calculator RR p.10 · Q9

[Maximum mark: 2]

$3^{-q} \times \frac{1}{27} = 81$. Find the value of q.

164

No calculator RR p.4 · Q16

[Maximum mark: 5]

(a) $(2^{24})^{\frac{1}{2}} = p^4$. Find the value of p. [2]

(b) Simplify $\frac{q^2 + q^2}{q^{\frac{1}{4}} \times q^{\frac{1}{4}}}$. [3]

165

No calculator Practice

[Maximum mark: 5]

Solve these equations (write both sides as powers of the same base).

(a) $2^x = 32$ **(b)** $3^x = \frac{1}{27}$ **(c)** $9^x = 27$

(d) $4^{x+1} = 8^x$ **(e)** $25^x = \frac{1}{125}$

163 *RR p.10 · Q9*

$3^{-q} \times 3^{-3} = 3^4 \Rightarrow -q - 3 = 4$ **M1**

$\boxed{q = -7}$ **A1**

164 *RR p.4 · Q16*

(a) $2^{12} = p^4 \Rightarrow p = 2^3$ **M1** $\boxed{p = 8}$ **A1**

(b) $\dfrac{2q^2}{q^{\frac{1}{2}}}$ **B1 B1**$= \boxed{2q^{\frac{3}{2}}}$ **A1**

165 *Practice*

(a) $2^x = 2^5 \Rightarrow \boxed{x = 5}$ **A1**

(b) $3^x = 3^{-3} \Rightarrow \boxed{x = -3}$ **A1**

(c) $3^{2x} = 3^3 \Rightarrow \boxed{x = \dfrac{3}{2}}$ **A1**

(d) $2^{2x+2} = 2^{3x} \Rightarrow 2x + 2 = 3x \Rightarrow \boxed{x = 2}$ **A1**

(e) $5^{2x} = 5^{-3} \Rightarrow \boxed{x = -\dfrac{3}{2}}$ **A1**